Soft Comput (2018) 22:2973–2999
https://doi.org/10.1007/s00500-017-2549-z

METHODOLOGIES AND APPLICATION

Facial emotion recognition with transition detection for students with high-functioning autism in adaptive e-learning

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Published online: 10 April 2017
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Abstract Emotions deeply affect learning achievement. In the case of students with high-functioning autism (HFA), negative emotions such as anxiety and anger can impair the learning process due to the inability of these individuals to control their emotions. Attempts to regulate negative emotions in HFA students once they have occurred, subsequent regulation to HFA students is often ineffective because it is difficult to calm them down. Hence, detecting emotional transitions and providing adaptive emotional regulation strategies in a timely manner to regulate negative emotions can be especially important for students with HFA in an e-learning environment. In this study, a facial expression-based emotion recognition method with transition detection was proposed. An emotion elicitation experiment was performed to collect facial-based landmark signals for the purpose of building classifiers of emotion recognition. The proposed method used sliding window technique and support vector machine (SVM) to build classifiers in order to recognize emotions. For the purpose of determining robust features for emotion recognition, Information Gain (IG) and Chi-square were used for feature evaluations. The effectiveness of classifiers with different parameters of sliding windows was also examined. The experimental results confirmed that the proposed method has sufficient discriminatory capability. The recognition rates for basic emotions and transitional emotions were 99.13 and 92.40%, respectively. Also, through feature selection, training time was accelerated by 4.45 times, and the recognition rates for basic emotions and transitional emotions were 97.97 and 87.49%, respectively. The method was applied in an adaptive e-learning environment for mathematics to demonstrate its application effectiveness.

Keywords Students with autism · Mathematics e-learning · Human–computer interface · Emotion recognition · Emotional transition

Communicated by V. Loia.

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1 Introduction

Human emotional expressions have an important role in human interaction. Many researchers in the emotion recognition field have attempted to improve sensitivity in human–computer interaction (HCI). Emotion is expressed in many different forms, including though facial expressions, body language, and vocalizations; among these, 55% of emotional information is conveyed by facial features (Mehrabian and Wiener 1967). Some studies have confirmed the potential for using facial expression to identify human emotions (Kapoor et al. 2007; Song et al. 2008; Senechal et al. 2012; Janssen et al. 2013; Bailenson et al. 2008; Tariq et al. 2012; Cowie et al. 2001).

Recently, many methodologies have been proposed for increasing the learning performance of students with disabilities (Ho 1999). With advances in information and communication technologies, e-learning has proven to be an effective learning method for students with disabilities. Computer-delivered methods also improve predictability and enable repetition of both activities and entertaining anima-

tions (Herskowitz 2000). High-functioning autism (HFA) students have been found to be more successful with regard to using e-learning as compared to students with other types of disabilities. HFA students also are deficient in social communication skills, and they have narrow repetitive interests (American Psychiatric Association 2000) as well as dispositions that are resistant to change (Wainer and Ingersoll 2011). Therefore, e-learning environments may meet both their behavioral and cognitive requirements (Swezey 2003; Cheng and Ye 2010), stimulate their learning motivation, and may also resolve many difficulties they experience during the learning process (Vullamparthi et al. 2011). Nevertheless, the core feature of HFA students is impaired emotion. Emotional reactions typically intensify (Reaven 2009), and frustration and impatience elicit negative emotions such as anxiety and anger and lead to interruptions and negative motivation toward learning progress. Regulating HFA students after emotions are elicited is ineffective, and calming an HFA student who has had negative emotions aroused is difficult. Also, teachers generally are located remotely, which makes difficult to regulate or decrease negative reactions on a timely basis. These shortcomings limit the effectiveness of e-learning. Therefore, the identification and delivery of assistive therapeutic materials regulate negative emotions in e-learning environments are necessary.

However, two main issues that are involved in recognizing emotion in HFA students in e-learning environments must be addressed. First, facial expression-based emotion recognition is a classification problem, and the conventional approach is to use public facial expression databases (such as GEMEP-FERA) (Bänziger et al. 2012; Valstar et al. 2011) to train the classifiers, but the facial expressions of HFA students change less than those of typical people (Bieberich and Morgan 2004; Czapinski and Bryson 2003), and studies have also shown that HFA students display unique patterns in regard to facial affect expression as compared to typical students (Yirmiya et al. 1989). Therefore, using existing databases to build classifiers for recognizing emotion is not suitable for them because these databases are based on typical humans. Second, in practice, emotion regulation in HFA students depends on emotion transition detection and recognition intended to provide a forewarning of negative emotions and to allow immediate preventive regulation. Despite the many emotion recognition methods that have been proposed, these methodologies only detect the results of emotion transitions. In addition, it is difficult for other people to recognize the forewarning of emotions of the face (Granhag and Strömwall 2004). Therefore, detecting a forewarning of emotion is more difficult than detecting emotions in general. Emotional preventive regulation is required not only in specialized e-learning domains. Any intelligent system of human interaction requires the capability to meet requirements related to emotion prevention, such as humanoid robots, intelligent agents (Becker et al. 2004).

The main objective of this study is to develop a facial expression-based emotion recognition detection method using transition detection, which includes two stages: classifier construction and recognition. Classifier construction is a process for training emotion classifiers based on facial feature data that are captured by experiments, and recognition is performed by using a previously constructed classifier to identify emotions with applications in real time. This study achieved this objective using the following steps: (i) an emotion recognition method with transition detection capability was developed by using a sliding window sampling technique to acquire related ensemble features for the purpose of training classifiers of emotion recognition; (ii) a Latin square-based experimental design was used to elicit emotion in an e-learning environment, and (iii) the effectiveness of the emotion recognition classifier was assessed, including basic emotion recognition and transition detection. The proposed methodology is applicable not only for e-learning by HFA students, but also in other domain applications by changing the experimental subjects and emotion elicitation materials.

The remainder of this study is organized as follows: the next section reviews related works classifying student emotional states and supporting applications. Section 3 introduces the proposed emotion recognition method with transition detection, including the facial features extraction and classifier. Section 4 describes how the emotions of HFA students were experimentally elicited in an e-learning environment for collection of emotion sample data, and then the experimental results of the emotion recognition method are provided in Sect. 5. Section 6 summarizes the findings of the study. Finally, conclusions are given in Sect. 7.

2 Related work

Many studies of facial-based emotion recognition have been conducted including only facial feature-based methods and multimodal methods, which are summarized in Table 1. In Tariq et al. (2012), facial features were extracted by incorporating Hierarchical Gaussianization (HG), Scale-Invariant Feature Transform (SIFT), and Optic Flow (OF), with SVMs to implement the recognition of emotion and person identification, reaching an 80% emotion recognition rate. In Rivera et al. (2013), a local directional number pattern (LDP) with an SVM classifier was used to recognize emotion in static images. A recognition rate of 99.1% was achieved by Cohn-Kanade (2000) in an emotional image database. In Senechal et al. (2012), the use of feature extraction by Local Gabor Binary Patterns achieved an 83.5% recognition rate.

Some multimodal approaches include (Bailenson et al. 2008), which used face landmark software to track fea-

Table 1 Review of seven studies of emotion recognition based on facial features

References	Feature type	Sampling from	Sampling method	Method	Perf.
Bailenson et al. (2008)	Facial +physiological	Slicing signal, 1 s		SVM, LogitBoost	95%
Kapoor et al. (2007)	Facial + postural + physiological + mouse behavior	Slicing signal, 150 s	Template tracking (Kapoor et al. 2002)	SVM + Gaussian process	79.17%
Song et al. (2008)	Facial + speech	Frame by frame		Tripled hidden Markov model, K-means	93.24%
Senechal et al. (2012)	Facial only	Single image,	LGBP	SVM	83.5%
Tariq et al. (2012)	Facial only	Clips of a couple of seconds	HG, SIFT, OF	SVM	80%
Rivera et al. (2013)	Facial only	Single image,	LDP	SVM	99.1%
Janssen et al. (2013)	Facial + speech + physiological	Sliding window, 10-sec step with 10-s window		SVM, MLP	78%

ture points and coordinate with physiological signals in order to recognize emotion on a second by second basis, using SVMs and the LogitBoost algorithm to achieve a 95% recognition rate on average individual subjects using SVMs. In Kapoor et al. (2007), a template-based facial landmark tracker (Kapoor et al. 2002), a posture sensor, skin conductance, and mouse behavior were used to detect frustration in gaming through averaging 150 s of feature variations, reaching a 79.17% rate of recognition using SVMs + Gaussian process classification. In Song et al. (2008), emotion recognition was performed by using a facial animation subset and speech features and by using a tripled hidden Markov model and a k-means algorithm to actualize multimodal emotion recognition, achieving a 93.24% recognition rate.

Unlike the approaches mentioned above, some cognitive-based recognition approaches were proposed (Ogiela 2013, 2010, 2013). In Ogiela (2013), a cognitive-based recognition was performed, using the syntactic analysis method to match the expertise-based expectation model and semantic patterns which were generated from source image, and reaches 90.9% recognition rate in recognizing the localization pathology types in medical images. However, to reflect on the presented study, it is difficult to build an expectation model in our research; the combination of facial feature set and emotional states is numerous and unspecific. Furthermore, to compare with typical student, the facial emotion expression of HFA students presented in unique patterns (Yirmiya et al. 1989). Therefore, to apply cognitive-based methodology in this study is difficult.

To sum up, these related works confirmed the potential effectiveness of facial-based emotion recognition. Notably, experiments performed in Janssen et al. (2013) showed that machine recognition of emotions outperformed human recognition of emotions. However, as noted above, detecting emotion transition is important. A common feature of the above-referenced state-of-the-art methods is slicing the signal into fixed time intervals and given emotional states for each signal segment. However, since emotion variance is a streaming signal, we argue that the time dependency of a signal should be considered to benefit detection of emotional transitions. Some researchers have shown the sliding window algorithm to be effective for processing streaming signal data in applications such as stock trend analyses (Jang et al. 1993) and physiological signal processing (Pan and Tompkins 1985). Janssen et al. (2013) also applied a sliding window algorithm to recognize emotions and achieved a 78% recognition rate with MLP and SVM classifiers, thus proving the possibility of using sliding windows in emotion recognition. In contrast, given their outstanding classification performance, SVMs are now widely used for emotion recognition. Therefore, in this study, SVM classifiers were used combined with the sliding window algorithm to recognize emotions.

3 Method of emotion recognition with transition detection

The proposed method has two phases: Recognition Classifier Building and Online Recognition (Fig. 1). The Recognition Classifier Building phase first constructs a classifier by using training data collected during the elicitation experiment. In the emotion elicitation experiment, facial landmark tracking is performed using the FACEAPI to track landmark points obtained by webcam and entrusted experts manually tag emotional states at fixed time intervals. Automated labeling is used to convert emotional states to emotion labels with two categories: basic emotion (BE) and transition emotion (TE). The Online Recognition Phase uses the previously built classifier to recognize the user emotions in applications such as emotion regulation in e-learning, human–computer interaction. The following subsections describe the main components, which include sliding window sampling, automated labeling, feature extraction, and classifier construction.

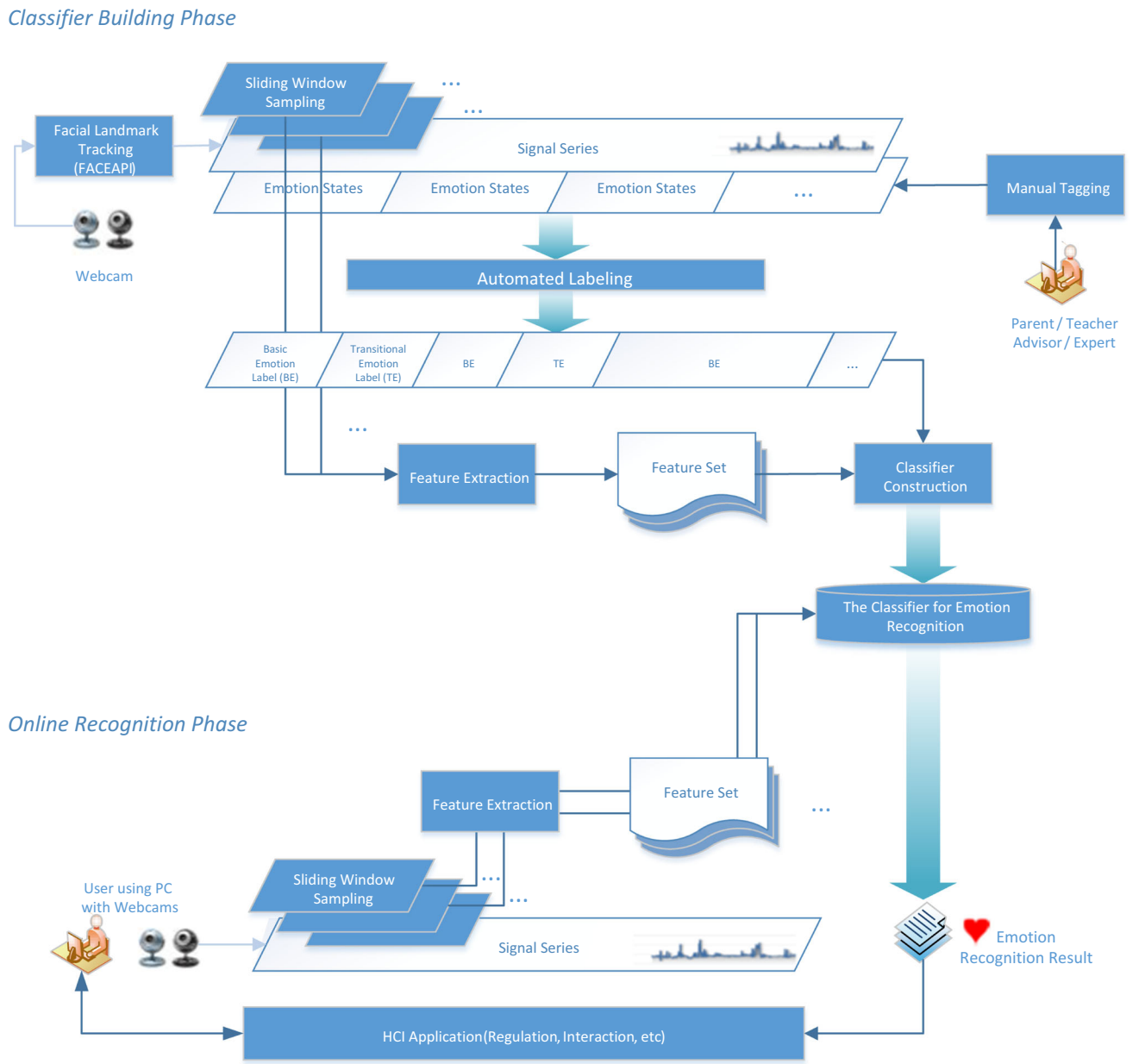


Fig. 1 Proposed method of emotion recognition with transition detection

3.1 Sliding window sampling

The goal of this step is sampling data from signals using a sliding window, which is composed of changes in facial landmark points. The two main parameters of the sliding window algorithm are window size and stepping (the distance between each window). This study matched the window size with a time interval for manually derived emotional states each time from the emotion elicitation experiment (for details of the experiment, see Sect. 4). The stepping is performed by using a tunable parameter, which significantly affects performance and accuracy. When stepping is increased, the recognition resolution decreases. Conversely, increasing the number of windows increases the number of corresponding samples and, consequently, increases the computational complexity.

The FACEAPI (Zintel 2008), a commercial function library for tracking and locating facial landmarks, was used for frame-by-frame capture of facial coordinates, which are independent of head pose. The webcam delivered 15 fps, 720*480 video streams to capture the changes in landmarks. Figure 2 shows the combination of facial landmarks defined by FACEAPI, with a total of 21 points. These points included 5 for each eye, 3 for each eyebrow, 4 for the mouth, and 1 for the nose.

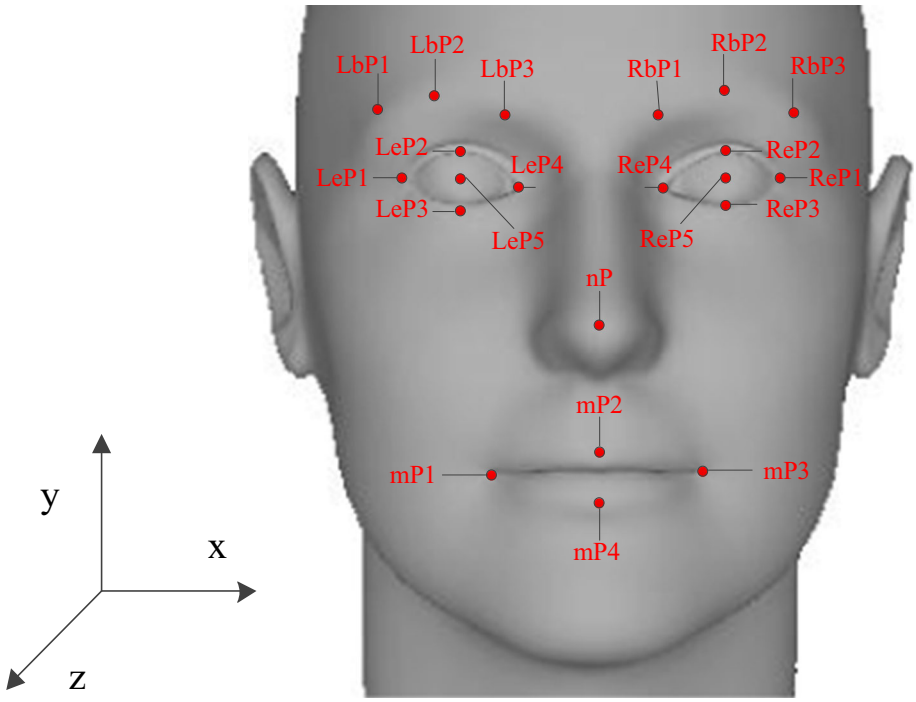


Fig. 2 Facial landmark distribution

Table 2 Emotion label list

Basic emotions (BEs)	Transitional emotions (TEs)		
Calm	Calm->happy	Calm->anxious	Calm->angry
Happy	Happy->calm	Happy->anxious	Happy->angry
Anxious	Anxious->calm	Anxious->happy	Anxious->angry
Angry	Angry->calm	Angry->happy	Angry->anxious

3.2 Automated labeling

Table 2 shows the emotion categories in this study: basic emotion (BE) and transitional emotion (TE). The BE is easily identified by humans and is easily identified in the face. Conversely, TE can be considered a forewarning of BE, which is difficult to identify in humans. In this study, an attempt was made to detect 4 BEs: calm, happy, anxious, and angry, and another 12 types of TEs were derived (Table 2). Figure 3 shows how automated labeling "emphasizes" a specific window, which may have TE-related ingredients within signals. The threshold is defined as the ratio of two emotions in a window (as in Fig. 3, *the green area/the orange area*). When the sliding window is moved over the border of the emotional state and reaches the threshold, the sample is labeled TE, as in Fig. 3. A sample labeled "B->A" denotes a transition from Emotion B to Emotion A. However, if the window has only one emotional state, the emotional state is still BE. We assume that these samples labeled by TE implicated ingredients which were related to transitions in emotion, and if the classifier could recognize the TE sample, detection of the emotion transition is made possible.

The main challenge of this process is determining the value of the threshold. The dilemma here is that, if the threshold is too low, the classifier may become too sensitive when detecting TEs; false warnings of emotion transitions may occur frequently in the online recognition stage. In contrast, an overly high threshold obtains samples that do not represent the overall variance in the samples in regard to emotion transition and makes accuracy lower. Also, in the context of learning, the sensitivity of negative emotions is higher than that of positive emotions. Considering all previously mentioned issues, in this study, the threshold was set to 0.25 of negative TEs (any angry emotions and any anxious emotions) and 0.50 of nonnegative TEs (any happy emotions and any calm emotions) in order to balance sensitivity and accuracy. The results showed that a lower negative TE threshold increases the sensitivity of negative emotion transition detection.

3.3 Feature extraction

The objective of feature extraction is to generate a feature set from previously captured face landmark signals from slid-

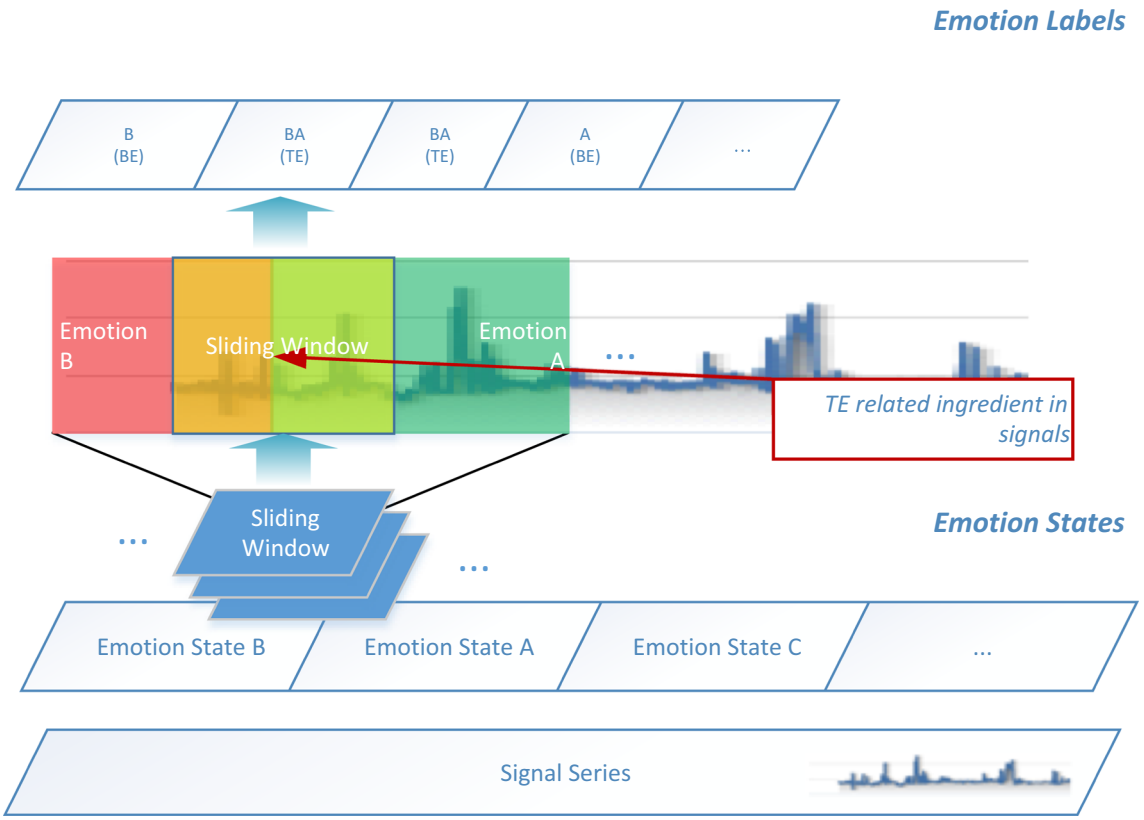


Fig. 3 The automated labeling process

ing window sampling (Fig 4). The three steps of feature extraction are landmark transformation, statistical processing, and normalization. The resulting feature set is then used to train samples of classifiers. The details of these actions are described as follows:

3.3.1 Landmark Transformation

For judging facial expressions, distances and angles were defined between each landmark to serve as observations of a facial expression (Fig. 5). Therefore, this step transformed all landmarks to distances and angles. Twelve distance-based observations and five angle-based observations were calculated using the landmark coordinates. Equations (3) and (4) are examples of the calculation of distance D5 and angle A14:

$$D5 = \sqrt{(mP2x - mP4x)^2 + (mP2y - mP4y)^2} \quad (1)$$

$$A14 = \cos^{-1} \frac{\overline{(mP1nP)} \cdot \overline{(mP3nP)}}{\left|\overline{mP1nP}\right| \cdot \left|\overline{mP3nP}\right|} \quad (2)$$

where

$D5$ height of mouth opening
A14 angle of the nose and both side corners of the month
mP1 right corner of the mouth
mP2 the center point of upper lips
mP3 left corner of the mouth
mP4 center point of bottom lips
nP nose

3.3.2 Statistical Processing

To measure the signal variance of a window, the standard deviation and average of each observation change were calculated for each window. Overall, 34 [(12 distances + 5 angles) × 2 statistical observations] features were generated, and 68 × 21 feature vectors were composed from these features. Figure 6 is an example of feature D5.

3.3.3 Normalization

If one feature has a broad range of values, the classifier may be governed by this particular feature. Therefore, the range of all features must be normalized to [0, 1], to ensure that each feature contributes a consistency ratio:

$$x' \frac{x - \min}{\max - \min} \quad (3)$$

where

x' normalized value

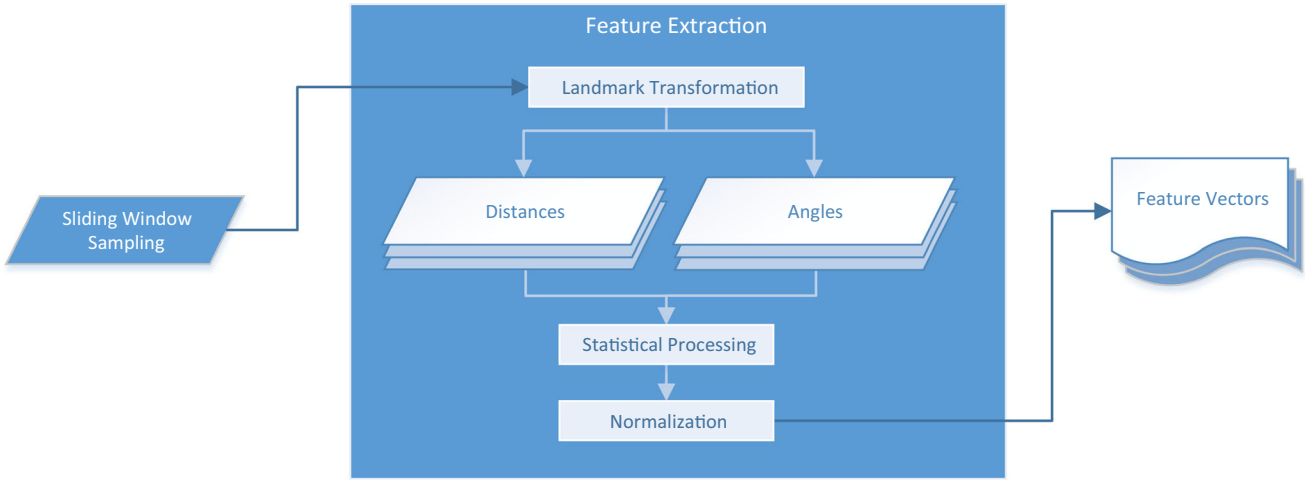
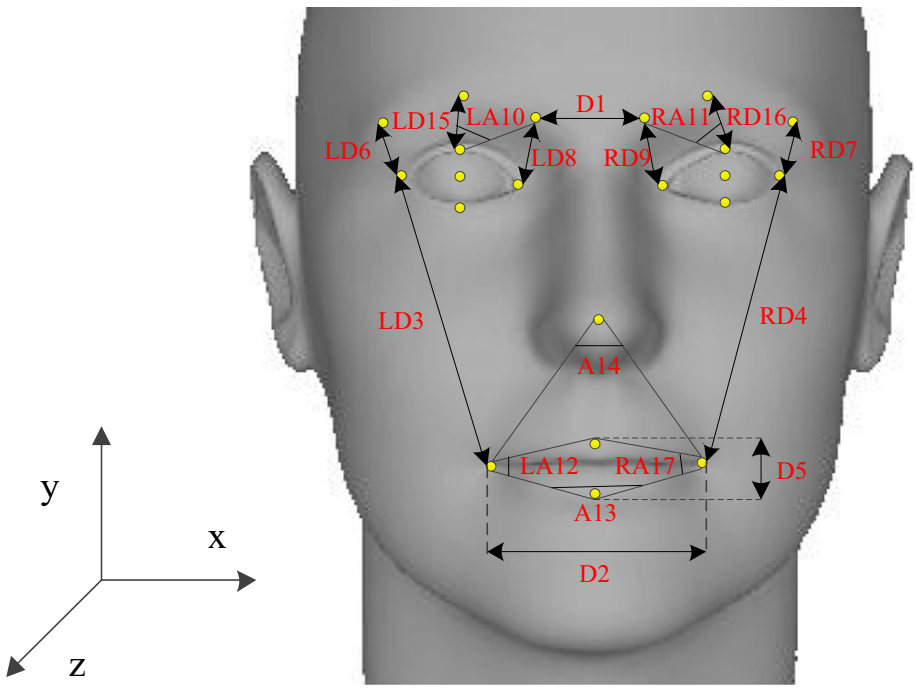


Fig. 4 Feature extraction flowchart

Fig. 5 Facial feature distribution



x original value of signals
min minimum value of original signals
max maximum value of original signals

Finally, the generated feature vectors can serve as training samples for classifier construction. One sample is denoted as a facial feature variation in the window, which consists of feature vectors with an emotion label. The next section describes how these training samples were used to build a classifier and to search for robust feature sets.

3.4 Classifier construction

This study used the SVM (Vapnik et al. 1996) technique to build classifiers, which were trained by previously generated feature vectors with emotion labels. Before training a classifier, feature selection is necessary. The aims of feature selection are to evaluate which features benefit the model construction, to evaluate which features lead off the transitional emotions, and to reduce the dimensions of the input feature vector to increase performance. As noted above, detecting a forewarning of human emotion is difficult. Therefore, identifying robust features associated with transitional emotions is important. Additionally, considering applications needs, real-time and rapid reactions in detecting negative emotion forewarnings are important, and reducing dimensions of vectors and focusing on robust features to decrease the response time and classifier building time is necessary (Figs. 7, 8).

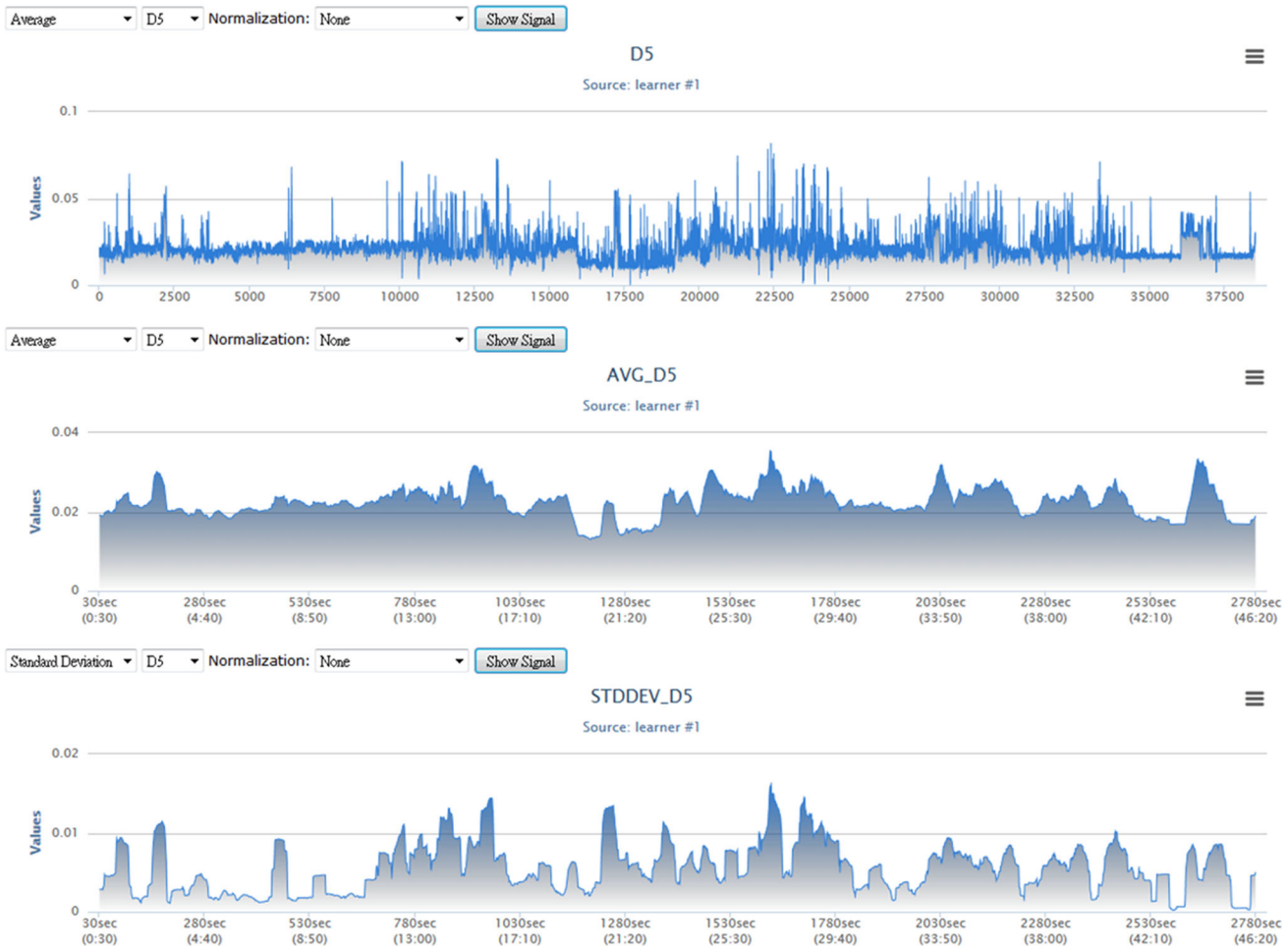


Fig. 6 Example of signal series (feature D5 with raw signal, averaged, and standard deviation)

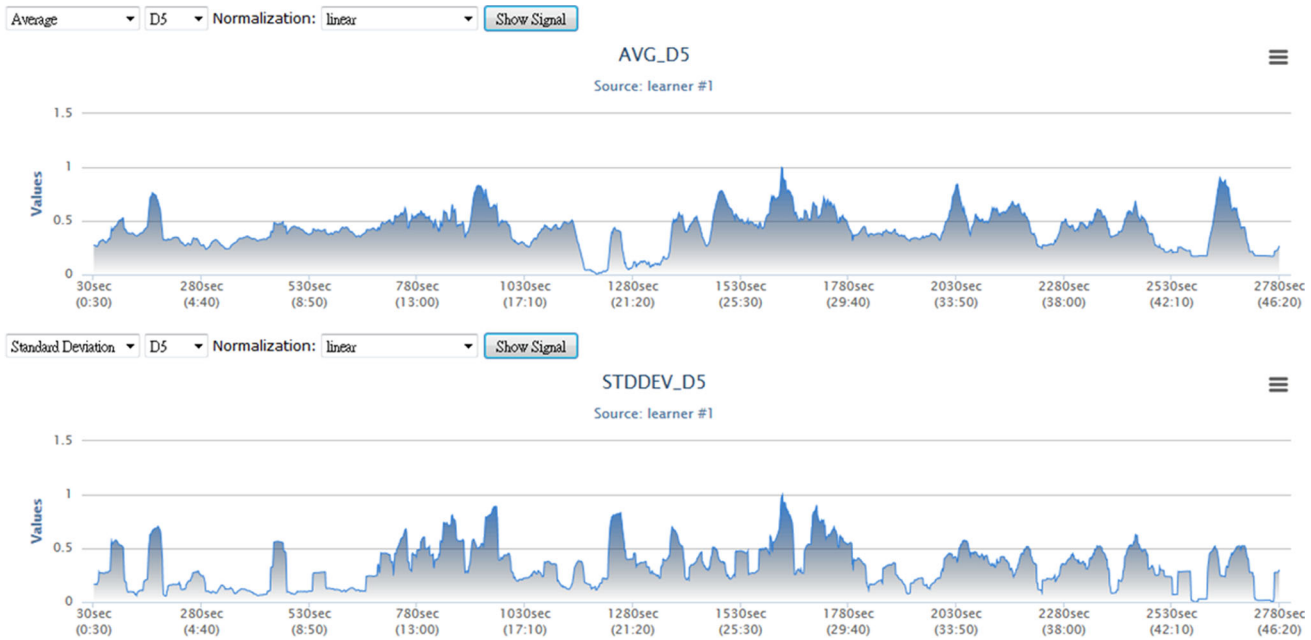


Fig. 7 Example of normalized signal series (feature D5)

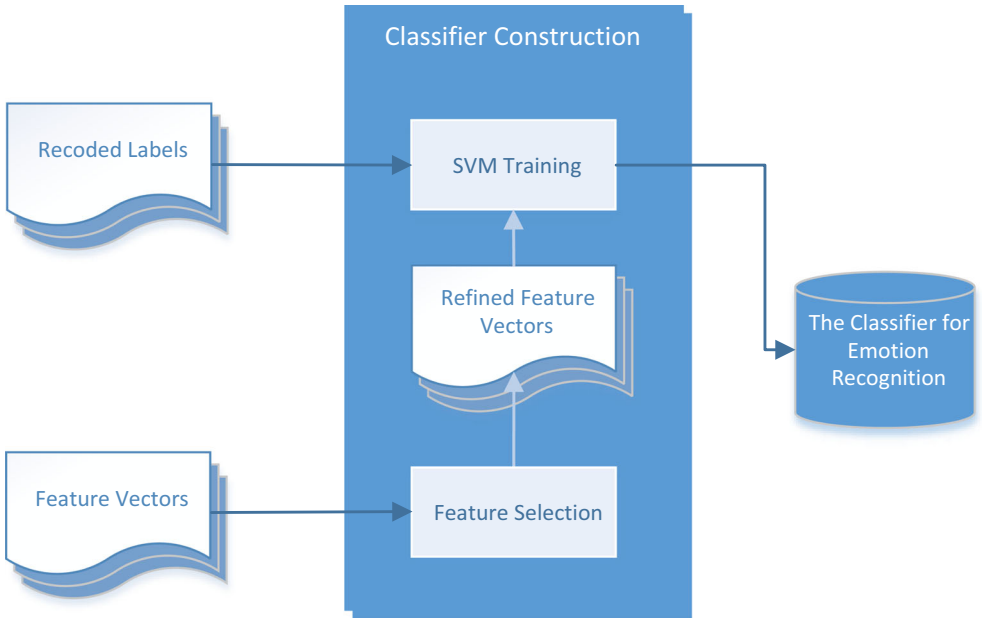


Fig. 8 Flowchart of classifier construction

Feature selection algorithms can be classified as filters and wrappers. In filter algorithms, features are first scored and ranked according to their relevance to the class label and then selected according to a threshold. Wrappers utilize a specific machine learning algorithm as a black box to score a feature subset by evaluating its predictive power. Since the filter approach is often computationally cost-effective, in this study, filters were used to search for and compare the optimized feature set. The Information Gain method (Quinlan 1979) was applied, and a Chi-square calculation was also used to rank each attribute.

Information Gain (IG) is a feature selection method (Quinlan 1979; Dhir et al. 2007) fundamentally defined as "the amount of information prior to testing" subtracted by "the amount of information after testing." As shown in Eq. (4), the amount of information a feature contains is calculated to judge whether the feature is selected. This was used to reduce the original feature sets to feature subsets that are easier to process, thereby reducing the feature dimension. This study uses IG to calculate the amount of information contained in each facial feature. "The amount of information before testing" corresponds to this study and represents the calculation of the total amount of information contained by the emotional categories. "The amount of information after testing" indicates the total amount of information contained in certain single features after information S is categorized. Larger IG indicates that the facial feature contains more information. These facial features are more important to the classification algorithm.

$$\text{InfoGain}(A_j) = \text{Info}(S) - \text{Entropy}(A_j) \quad (4)$$

where Info(s) total amount of information contained by emotion categories.

Entropy(A_j) total amount of information contained in a certain facial feature after information is categorized.

A_j A certain single facial feature.

Also, this study applied Chi-square feature selection to evaluate the contribution of each facial feature by computing the value of the Chi-squared statistic with respect to the emotion tagging, to find the most relevant facial features for the dataset.

In classifier building stage, the SVM classifier parameters (*cost C* and *gamma* γ) are tuned using a grid search method (Huang et al. 2007) to maximize the accuracy of the classifier parameter. A grid search is a model selection procedure. Through trying various pairs of (C, γ), the best result can be obtained. However, testing all combinations of (C, γ) is time-consuming, grid search uses coarse grids to search the region that may contain a better result first. After the region that contains better result is found, a finer search of this region can be conducted recursively. In this work, we found the best (C, γ) combination to be (C = 16.0, γ = 10.0).

4 Emotion elicitation experiment

This study simulated e-learning environments to elicit four different emotions in HFA students to collect face landmark data for the purpose of emotion recognition model construction: anger, anxiety, calmness, and happiness. Research

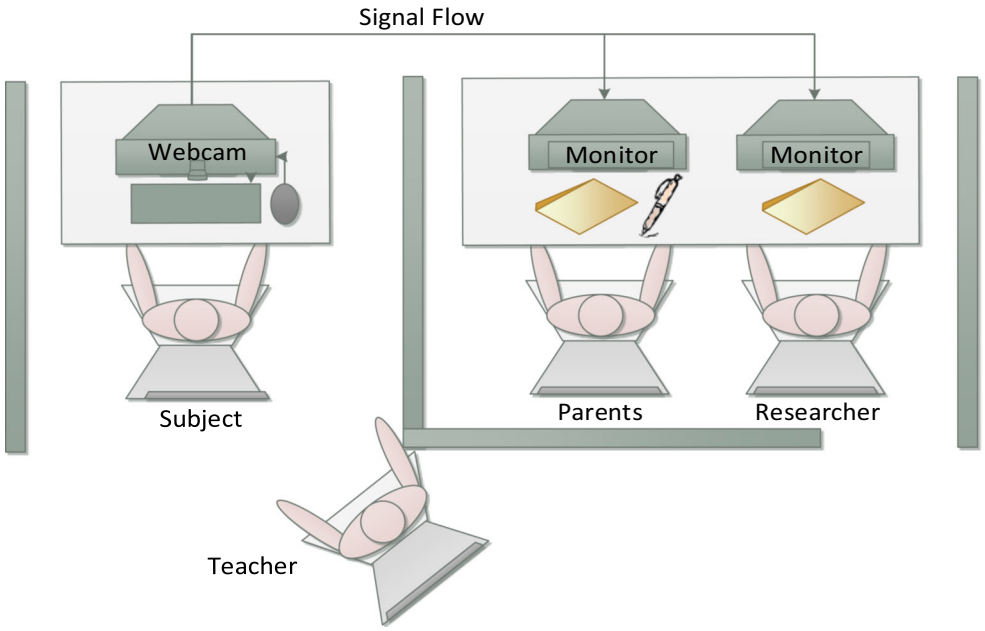
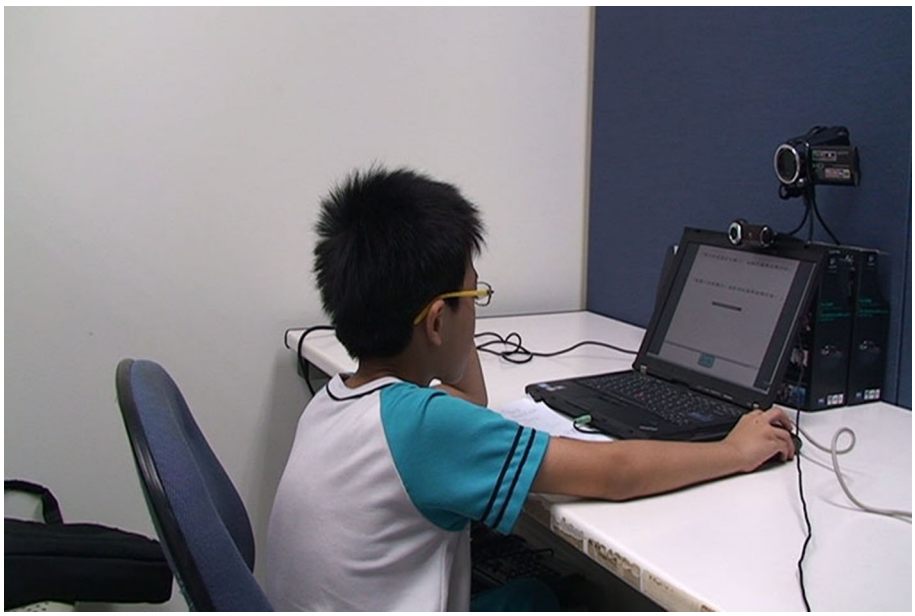


Fig. 9 Emotion elicitation experimental environment

Fig. 10 An HFA student performing the exercise



shows that mathematics is the most difficult subject for elementary and secondary school students (Mazzocco and Myers 2003). Therefore, this study elicited emotional reactions from mathematics learning. A webcam was installed to record changes in facial expressions. To avoid external interference when performing the experiment and to facilitate the smooth integration of the subjects into the learning environment, an experimental learning space was installed as an independent environment. Figures 9 and 10 show an overview of the environment used for the emotion elicitation experiment. During the emotional elicitation period, both the researchers and parents sat in front of a monitor, and the researchers controlled the transmission of facial feature signals at all times. Abnormal signals caused by equipment problems were minimized, and a special education teacher served as a facilitator to guide subjects into the learning scenario. If the subjects lost control of their emotions, or if unexpected situations occurred, the teacher was available to provide timely comfort and assistance. Parents and the teacher observed the changes of emotion and tagged each sample with an emotion state.

4.1 Subjects of emotion elicitation Experiment

The subjects were fifteen HFA students (age range 8–12 years); all but one child was male. All subjects were recruited from elementary school special education classes. Additionally, each subject had an independent clinical diagnosis of autism spectrum disorder that was confirmed with the DSM-IV-TR criteria for autism spectrum disorder (American Psychiatric Association 2000). Generally, the participants spent half of their time in a resource classroom and half of their time in mainstream education classrooms with special education staff support. Since this study used mathematics e-learning to elicit emotions, participants were required to have basic mathematical knowledge, language, and communication skills and the ability to use an e-learning system interface, to type on a keyboard and to use a mouse. Hence, for the purpose of the experiment, only participants with FSIQ/VIQ/PIQ scores above 70 were recruited. The FSIQ/VIQ/PIQ scores were obtained using the Wechsler Intelligence Scale for Children-III (WISC-III).

4.2 Materials for eliciting emotions

To elicit the four target emotions in this experiment, the authors consulted autism experts in special education field to design three mathematics e-learning contexts and calmness context. Figure 11 shows the materials used to elicit these emotions. The elicitation methods are described below:

Calmness In order to gain the attention of the subject, 20 different geometric shapes were used and connected. The shapes were transformed every 10 s, and the subjects were asked to calculate the total number of changes in shape or color. The extremely slow transformations had a calming effect on the subjects.

Happiness This emotion was elicited by e-learning materials such as mathematics games and humorous video clips. Subjects were allowed to choose the material according to their preferences.

Anger A curriculum-based examination was used as the learning material in this context, which included 18 questions progressing from low to high difficulty. To elicit

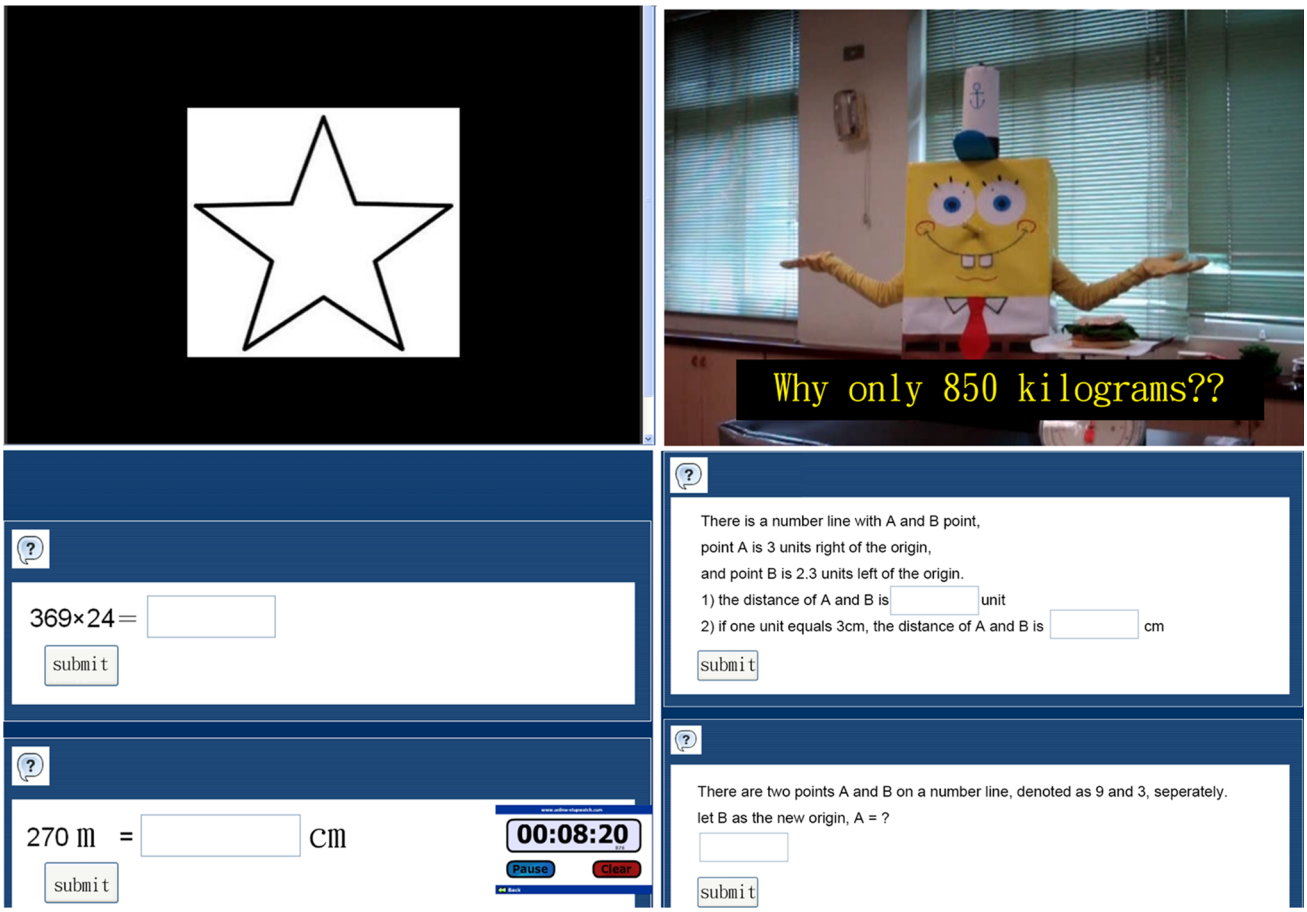


Fig. 11 Materials used to elicit emotions (clockwise from top left): (1) count graphic shapes and colors; (2) an interesting mathematical story; (3) difficult mathematical exercise; and (4) time-limited competition

Table 3 Emotion elicitation sequence

Round	Emotion elicitation sequence
1	Calmness → happiness → calmness → anxiety → calmness → anger
2	Calmness → anger → calmness → happiness → calmness → anxiety
3	Calmness → anxiety → calmness → anger → calmness → happiness

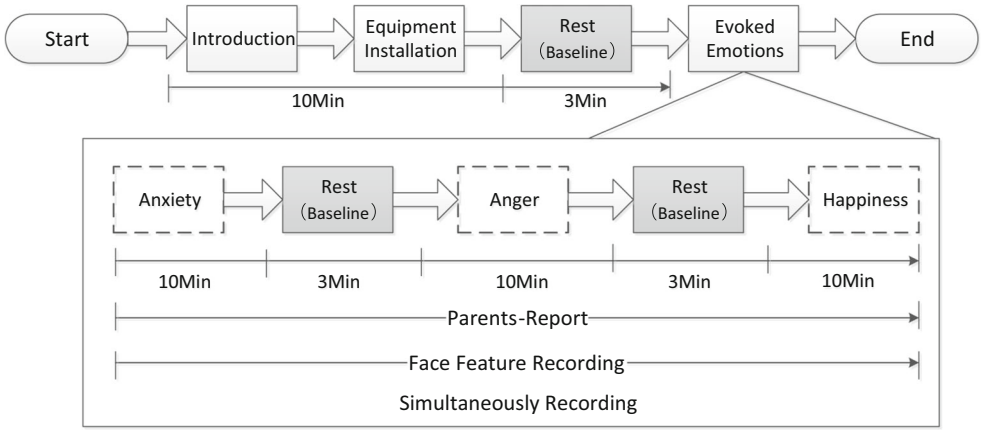


Fig. 12 Experimental process for eliciting emotions

angry reactions, the difficulty of the tests exceeded the capabilities of the participants.

Anxiety This context used a time-limited competitive math test to induce anxiety in the subjects. The content of the test was taken from a curriculum-based questions database, and the winner of the competition received an award. The test was designed to elicit anxiety by introducing sources of stress, such as the expectation of rewards, number of test items (50 items), and a time limit (10 min).

4.3 Experiment procedure

The instructor asked the subjects to participate in two-stage mathematics e-learning exercises. The first exercise was a pre-test intended to ensure that the nontargeted emotions were not induced and to acclimate the subjects to the experimental environment. The second exercise was the formal experiment. The Latin square experimental schema (Kirk 1995) applied in this study used partial counterbalancing to control the order effect (Hogarth and Einhorn 1992) and carry-over effects (Macfie et al. 1989). Table 3 shows the experimental scenario sequence, and Fig. 12 shows the implementation steps.

4.4 Manual tagging with emotional states

Compared to typical students, HFA students have more difficulty describing their own experiences during experiments, especially their moods and psychological states (Baron-Cohen 2000). Additionally, since they often have barriers to emotional communication, their self-assessments of the effectiveness of emotional stimulation are often inaccurate. Therefore, parents and experts accompanied the participants throughout the experiment to aid researchers with regard to judging the emotional states of subjects.

During the experiment, the expert teachers and parents were asked to complete a form indicating the emotional state of the subjects at that time, including angry, anxious, happy, and calm, with synchronized collected facial emotion signals elicited every 30 s. The combined tagging results of experts, parents, and emotion type were used to judge emotions elicited in each sample of signals. The final dataset included 1156 emotional states. Of these, the state was calm in 424, happy in 165, anxious in 460, and angry in 107. Each sample also included related facial landmark signals.

5 Evaluation

5.1 Evaluation protocol

5.1.1 Evaluation indices and methods of proposed methodology

The evaluation protocol, which included the evaluation procedure (Fig. 13) and evaluation indices, revealed three findings. First, the sliding window algorithm, which had two parameters, steps and windows size, had a large impact on resolution and performance. This work applied 1- and 5-s sliding window algorithm steps, and the results were

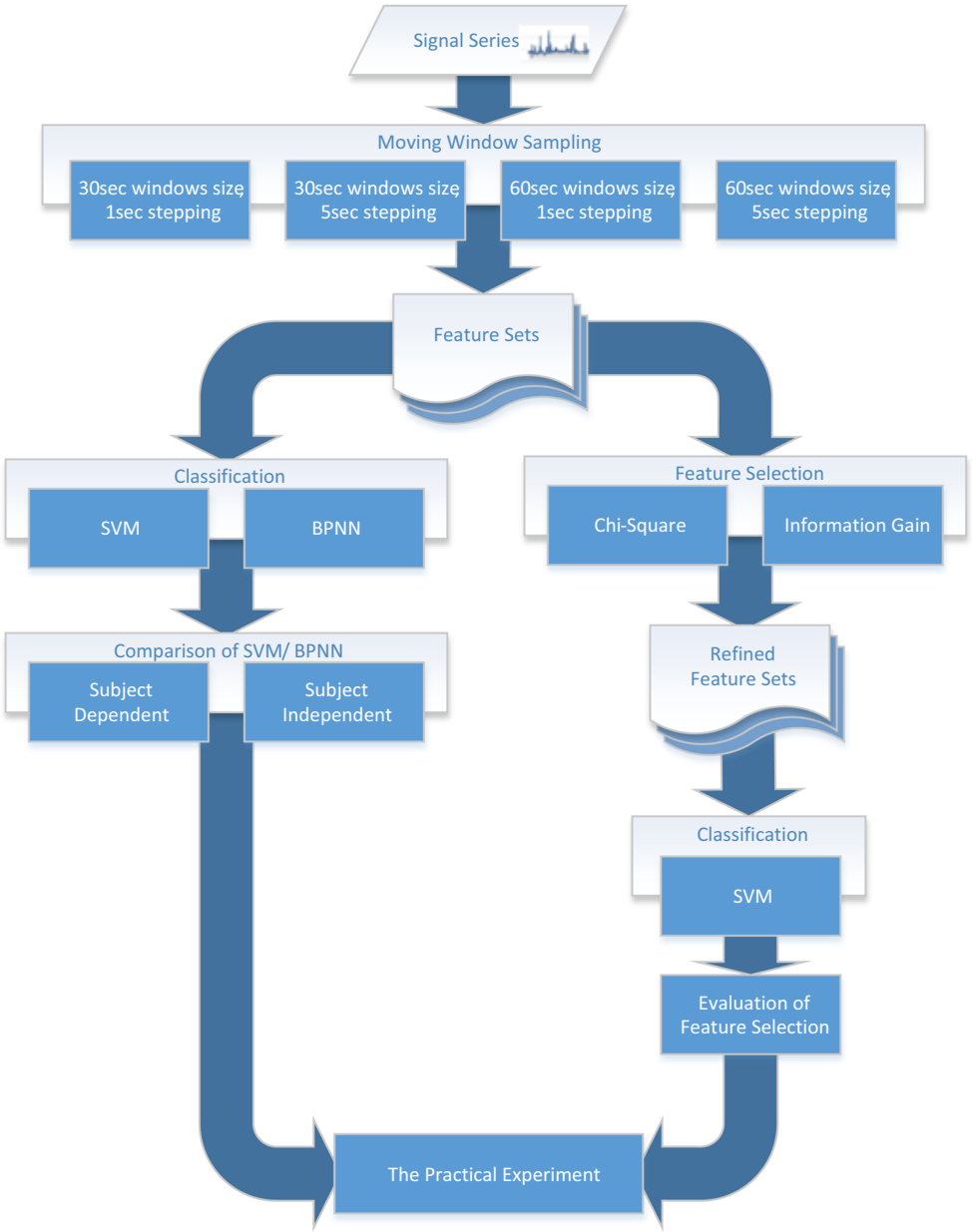


Fig. 13 The evaluation procedure

evaluated using comparative analysis, including subject-dependent analysis and subject-independent analysis. For alignment of emotional states, the window size should be configured as a multiple of 30 s. Therefore, this work used 30- and 60-s window sizes for the purpose of evaluation.

Second, in this work, as mentioned in Sect. 3.4, IG and Chi-square were used for feature selection to find relevant features of emotion recognition. Besides the evaluation of these methods, the effectiveness of sliding window configurations, including window size and stepping size, was investigated.

Third, in order to demonstrate the effectiveness of the proposed method, the BPNN (back-propagation neural networks), a widely used method in machine learning, was selected for comparison. The three parameters of the BPNN include learning rate, momentum, and the number of hidden nodes. This research configured BPNN using trial-and-error to search parameters (Ogiela 2013) and set the learning rate as 0.3 and the momentum as 0.2. The following rules were then used to determine the number of the hidden nodes (Liming 1993):

$$n_{\text{hidden}} = \log_2 N \tag{5}$$

where n_{hidden} is the number of hidden nodes, and N is the size of input vectors. To balance recognition performance and time cost, the BPNN ran 2000 epochs for large data (subject-

Table 4 Subject information

ID	Gender	Age	Type
1	F	10	Moderate ASD
2	M	9	Mild ASD
3	M	14	Mild ASD
4	M	8	Mild ASD
5	M	9	Mild ASD
6	M	9	Moderate ASD
7	M	9	Asperger
8	M	8	Asperger
Average age		**9.5**	
SD		**1.92**	

independent evaluation) in the training stage and ran 500 epochs for small data in the subject-dependent evaluation.

To verify the effectiveness of the classifiers, a tenfold cross-validation was also used to avoid over-fitting (Kohavi 1995). To compare the time cost and discrimination of the classifier, for searching an optimized parameter combination, the indices used in the evaluation protocol included Precision/Recall, F-Measure, ROC AUC, and Building Time, which is detailed as follows:

Precision/Recall and F-Measure this work used precision (P), recall (R), and the F-measure, which was defined as the harmonic mean between precision and recall, to assess performance in emotion classification using the following equations:

$$\begin{aligned} \text{precision} &= \frac{\{\text{relevant emotion} \cap \text{retrieved emotion}\}}{\{\text{retrieved emotion}\}} \\ &= \frac{\text{TP}}{\text{TP} + \text{FP}} \end{aligned} \tag{6}$$

$$\begin{aligned} \text{recall} &= \frac{\{\text{relevant emotion} \cap \text{retrieved emotion}\}}{\{\text{relevant emotion}\}} \\ &= \frac{\text{TP}}{\text{TP} + \text{FN}} \end{aligned} \tag{7}$$

$$\text{FMeasure} = \frac{2RP}{R + P}, \tag{8}$$

Table 5 Feature vector set information (unit: data size)

Win. size	Stepping	BEs Calmness	TEs →Happiness	→Anxious	→Anger	Total
30 s	5 s	2614	85	213	21	2933
	1 s	13,199	378	951	92	14,620
60 s	5 s	2351	168	397	38	2954
	1 s	11,822	778	1944	182	14726
		Happiness	→Calmness	→Anxious	→Anger	
30 s	5 s	995	113	10	5	1123
	1 s	5014	509	44	23	5590
60 s	5 s	858	226	18	9	1111
	1 s	4348	1049	90	45	5532
		Anxious	→Calmness	→Happiness	→Anger	
30 s	5 s	2849	112	28	111	3100
	1 s	14,373	503	124	489	15,489
60 s	5 s	2607	222	54	209	3092
	1 s	13,144	1028	253	1023	15,448
		Angry	→Calmness	→Happiness	→Anxious	
30 s	5 s	580	8	4	100	692
	1 s	2949	38	18	446	3451
60 s	5 s	471	16	8	179	674
	1 s	2383	74	36	868	3361
			Total	30 s	5 s	7848
					1 s	39,150
				60 s	5 s	7831
					1 s	39,067

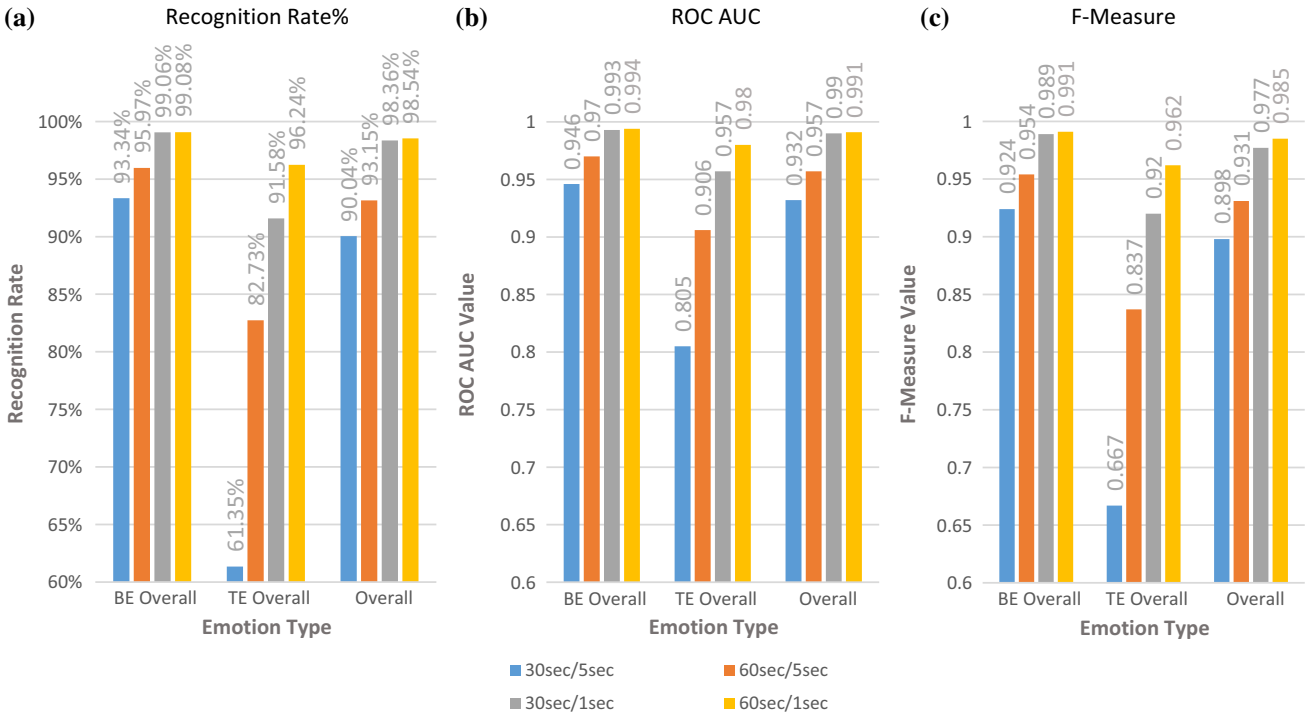


Fig. 14 **a** The average recognition rate of overall emotion type for subject-independent result (*left*), **b** the ROC AUC value of overall emotion type for subject-independent result (*center*), **c** the F-measure value of overall emotion type for subject-independent result (*right*)

where

TP Number of emotion categories classified correctly.
FP Number of emotion categories classified incorrectly.
FN Number of samples belonging to certain emotion categories but classified incorrectly.

ROC (receiver operating characteristic) AUC (area under curve) The ROC curve is a standard technique for summarizing classifier performance over a range of trades using true-positive and false-positive error rates (Swets 1988). The ROC curve is a robust method of identifying potentially optimal classifiers (Provost and Fawcett 2001). An ROC AUC over 0.7 indicates that the classifier has sufficient discernibility.

Classifier building time When training a large dataset, building classifiers are time-consuming. Therefore, the dimension size of the dataset must be reduced by feature selection. Hence, measuring the time cost of classifier building could be used to evaluate the performance of feature selection method. To evaluate classifier building time, this work used a PC with an Intel Xeon E3-1230V2 Processor, 16 GB RAM, and the Windows 7 operating system.

5.1.2 Effective evaluation in empirical experiment

To demonstrate the effectiveness and importance of detecting transitional emotions, this study applied proposed emotion recognition mechanism in real e-learning environment. Subjects were asked to attend high stress mathematics practice contests in two scenarios, *Time Limit Contest Scenario* and *High Difficulty Learning Scenario*. The e-learning platform showed treatment material for emotional regulation when the negative transitional emotions of the subjects (any anxious emotions and any angry emotions) were detected. Same as elicitation experiment (mentioned in Sect. 4), expert teachers in special education were invited to observe the effectiveness of the emotion recognition and regulation from two perspectives: the rate of negative behaviors and learning performance. The former stands for the ratio of negative behavior occurring over the whole experimental process. When the emotion of the subject was negative, the negative behavior tended to happen frequently. The latter referred to the number of correct answers per minute. The whole process was conducted in two stages: baseline stage and alternative stage, which stands for "without" and "with" emotion regulation. Table 4 lists the demographic information of the participants.

The *Wilcoxon signed rank test* has been proven to be representative for small sample sizes (Siegel 1956). Since the proposed empirical experiment was a small sample analysis, this method was used for evaluating noticeable differences with and without the transitional emotion detection with regulation strategy interventions to verify the feasibility and effectiveness of this study.

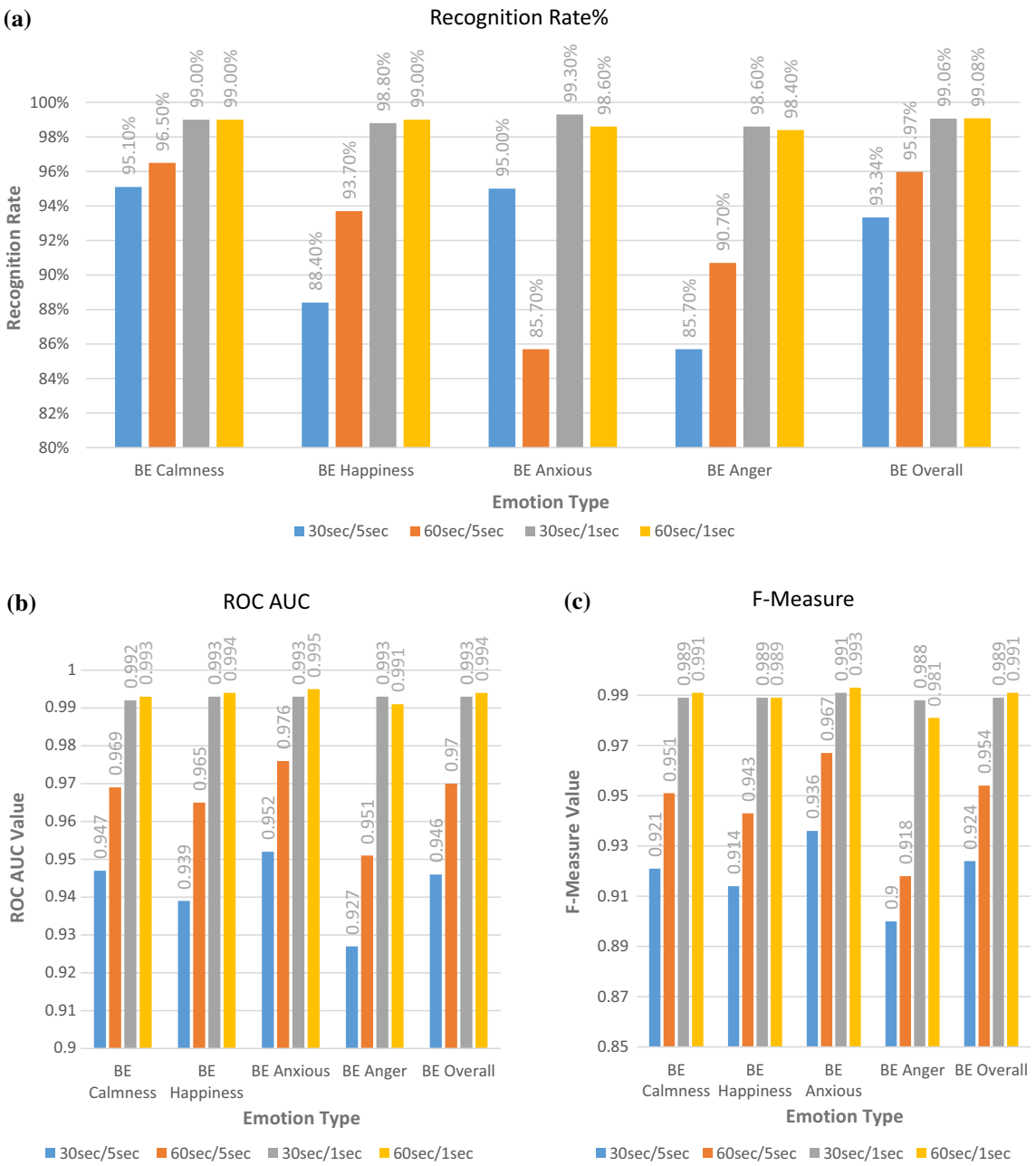


Fig. 15 **a** Average recognition rate of basic emotion type for subject-independent result (*top*), **b** ROC AUC value of basic emotion type for subject-independent result (*bottom-left*), **c** F-measure value of basic emotion type for subject-independent result (*bottom-right*)

5.2 The result of proposed methodology

5.2.1 Feature vector set

As described in Sect. 4.4, 1156 emotion states were collected from the experiment. Sliding window sampling (Sect. 3.1) and automated labeling (Sect. 3.2) were used to transform facial landmark signals into feature vectors. Because of the different stepping settings, the sliding window resulted in different increases in the size of the feature vector set, and the size of the feature vector set, as a classifier training sample, had a big impact on training time and accuracy. Table 5 shows the feature vector set information. Compared with the 5-s stepping sliding window sampling, the 1-s stepping setting increased the feature vector set substantially.

5.2.2 Performance evaluation

Figures 14, 15, and 16 show the performance indices of various step and window size settings. Regardless of the window

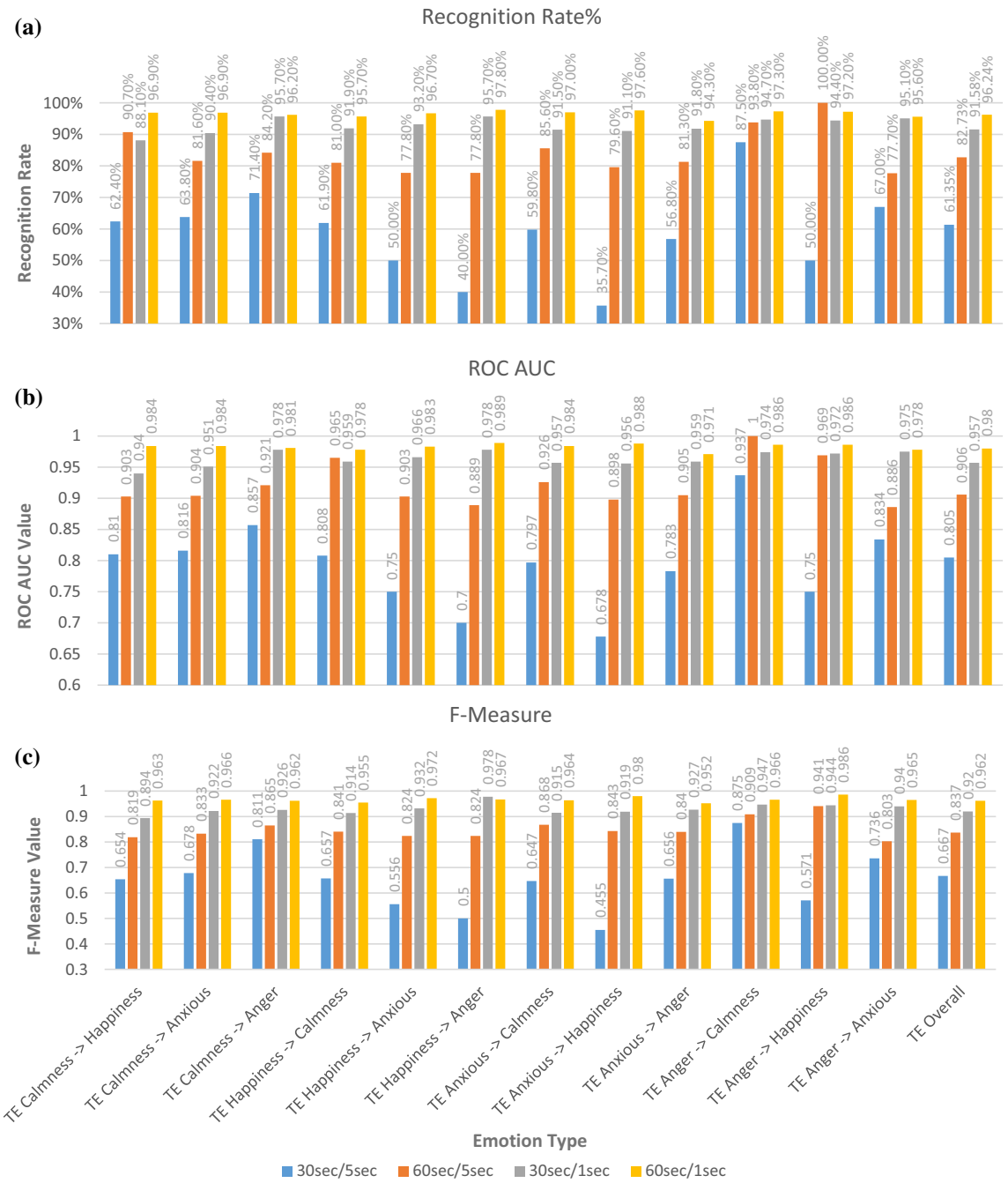


Fig. 16 **a** Average recognition rate of transitional emotion type for subject-independent result (*top*), **b** ROC AUC value of transitional emotion type for subject-independent result (*middle*), **c** F-measure value of transitional emotion type for subject-independent result (*bottom*)

size, the 1-s stepping obtained substantially better results as compared to the 5-s stepping. In the subject-independent evaluation, in Fig. 14, the average overall recognition rate was 98.36/98.54% in 30-/60-s window size, respectively, but that of the 5-s stepping was only 90.04/93.15% in 30-/60-s window size, respectively. Although no significant improvement in BE detection in Fig. 15, 1-s stepping substantially improved recognition power by increasing the sampling resolution in TE detection. It reached 91.58/96.24% in the 30-/60-s window size, respectively. Conversely, 5-s stepping achieved a TE recognition rate of only 61.35/82.73% in the 30-/60-s window size, respectively. The TE-related results are shown in Fig. 16.

Also, results displayed in the larger window size were more useful with regard to recognizing TE detection. In 5-s stepping, TE detection improved from 61.35 to 82.73%,

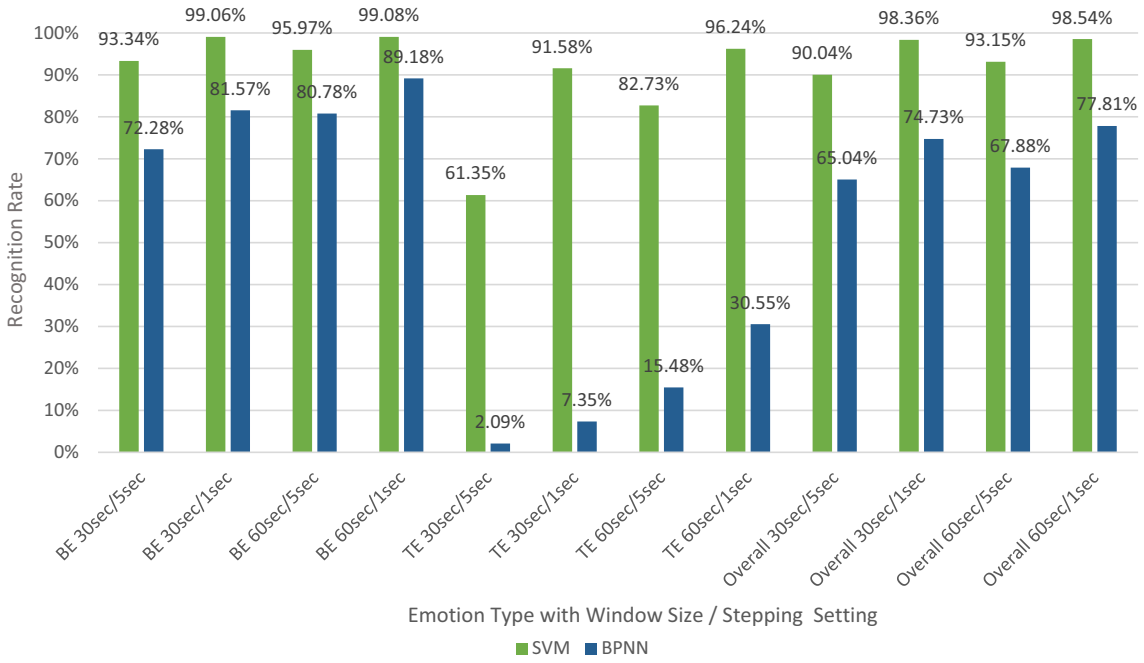


Fig. 17 Average recognition rate of emotion type for subject-independent comparison result of BPNN/SVM

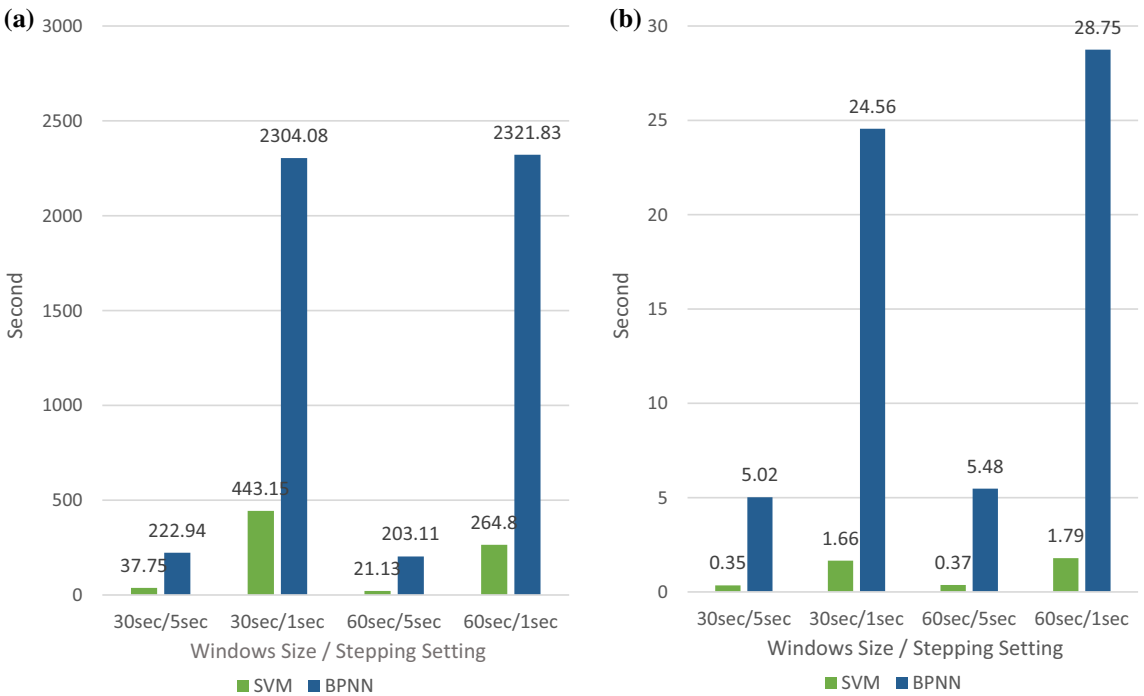


Fig. 18 **a** Classifier building time of subject-independent comparison result of BPNN/SVM (*left*), **b** average classifier building time of subject-dependent comparison result of BPNN/SVM (*right*)

and in 1-s stepping, TE detection improved from 91.58 to 96.24%. However, each setting provided a certain level of discernibility. All ROC AUC values exceeded 0.7.

The analytical results of the comparison of BPNN and SVM are shown in Figs. 17, 18, and 19. The subject-dependent evaluation also obtained similar results in the two stepping settings with different window sizes. The 1-s stepping setting achieved higher average recognition rates as compared to the 5-s stepping settings (Fig. 19, red line and sky blue line), and the larger window size improved

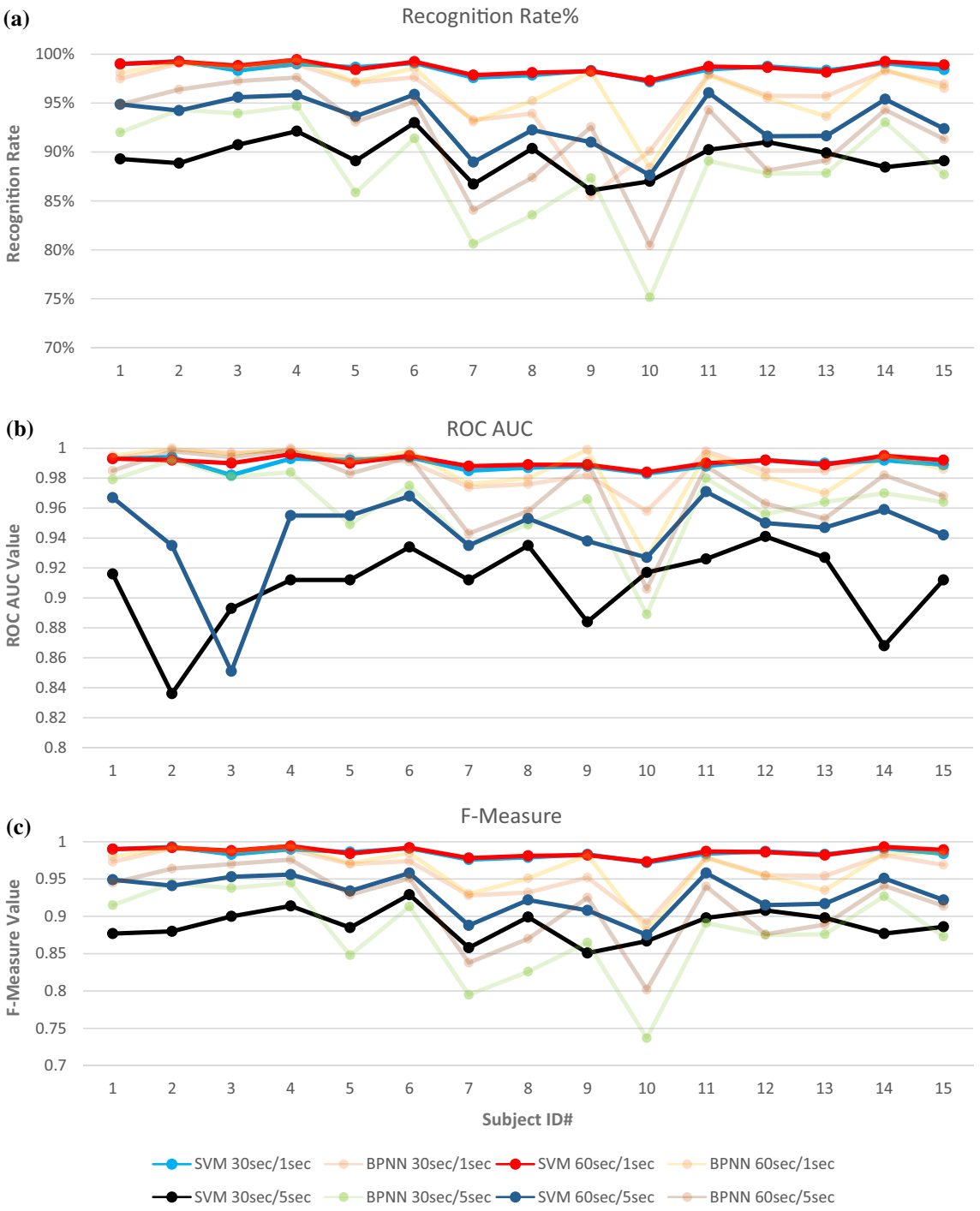


Fig. 19 **a** Average recognition rate of subject-dependent comparison of BPNN and SVM (*top*), **b** ROC AUC value of subject-dependent comparison of BPNN and SVM (*middle*), **c** F-measure value of subject-dependent comparison of BPNN and SVM (*bottom*)

recognition performance. Compared with the BPNN, no substantial difference was discovered in the subject-dependent evaluation. However, better discriminability was obtained in subject-independent evaluation (Fig. 17). In addition, from a time-cost perspective, compared with the SVM, the BPNN was found to be time-consuming (Fig. 18).

5.2.3 The evaluation result of feature selection

Tables 6 and 7 show the evaluation results for feature selection with different window size settings and stepping settings. To evaluate the effectiveness in reducing dataset dimensions, this work used 3 subdatasets with features sizes reduced by

Table 6 The ranking of relevant features—with 30-s window size

#	Step size: 1 s						Step size: 5 s						Reduced by
	Chi-square			Information Gain			Chi-square			Information Gain			
	Name	SO	Score	Name	SO	Score	Name	SO	Score	Name	SO	Score	
1	RA11	Avg.	18418.46	D2	Avg.	.154	A14	Avg.	1333.69	D2	Avg.	.111	
2	RA11	Std.	11729.76	A14	Avg.	.152	RA11	Avg.	130.61	A14	Avg.	.109	
3	LA10	Avg.	11461.59	RA11	Avg.	.128	D2	Avg.	1266.79	RD4	Avg.	.092	
4	A14	Avg.	10727.71	RD7	Avg.	.123	RD4	Avg.	1164.82	A13	Avg.	.090	
5	D2	Avg.	10608.59	LD6	Avg.	.123	D5	Avg.	116.43	RA11	Avg.	.087	
6	LD3	Avg.	8721.82	LA10	Avg.	.121	A13	Avg.	1007.62	D5	Avg.	.083	
7	D5	Avg.	8615.57	A13	Avg.	.120	LA10	Avg.	972.01	LD3	Avg.	.078	
8	RD7	Avg.	8488.11	RD16	Avg.	.120	LD3	Avg.	838.17	LD6	Avg.	.074	75%
9	RD16	Avg.	8112.32	RD4	Avg.	.119	RA11	Std.	789.50	LD15	Avg.	.069	
10	A14	Avg.	7866.80	LD3	Avg.	.116	LD6	Avg.	761.63	LA10	Avg.	.065	
11	D1	Avg.	7755.72	D5	Avg.	.180	LD8	Avg.	687.92	LD8	Avg.	.054	
12	RD4	Avg.	7267.96	LD15	Avg.	.101	LD15	Avg.	683.06	RA11	Std.	.053	
13	LD15	Avg.	7185.61	RD9	Avg.	.096	RD7	Avg.	526.71	RD7	Avg.	.053	
14	D5	Std.	7123.37	LD8	Avg.	.095	RD16	Avg.	499.02	RD16	Avg.	.049	
15	RA17	Avg.	7123.04	D1	Avg.	.093	D1	Avg.	498.32	D1	Avg.	.047	
16	A13	Std.	7041.90	RA11	Std.	.086	RA17	Avg.	489.13	RA17	Avg.	.036	
17	LD6	Avg.	6913.35	LA10	Std.	.080	LA12	Avg.	476.91	LA12	Avg.	.036	50%
18	LA12	Avg.	6586.85	D5	Std.	.077	LD3	Std.	440.38	D1	Std.	.033	
19	LD3	Std.	6527.51	LD3	Std.	.075	RD4	Std.	429.93	RD7	Std.	.032	
20	RD4	Std.	6447.63	RD4	Std.	.075	LD15	Std.	389.60	LA10	Std.	.031	
21	LD8	Avg.	6373.04	RD16	Std.	.075	A13	Std.	370.57	A13	Std.	.031	
22	RD9	Avg.	6265.58	RD7	Std.	.073	LA10	Std.	369.49	RD16	Std.	.031	
23	LA10	Std.	6229.58	A13	Std.	.073	RD7	Std.	365.93	LD3	Std.	.029	
24	RD16	Std.	5249.77	RA17	Avg.	.073	D1	Std.	345.85	RD4	Std.	.028	
25	RD7	Std.	5225.80	D1	Std.	.071	RD16	Std.	334.13	RD9	Avg.	.028	25%
26	D1	Std.	4777.81	LA12	Avg.	.064	D5	Std.	293.82	LD6	Std.	.025	
27	D2	Std.	4746.86	LD6	Std.	.064	RD9	Avg.	291.35	D5	Std.	.024	
28	LA12	Std.	4580.84	LD8	Std.	.060	LD6	Std.	270.14	LD15	Std.	.023	
29	LD15	Std.	4574.12	A14	Std.	.059	LD8	Std.	250.76	LD8	Std.	.022	
30	RA17	Std.	4484.92	D2	Std.	.059	D2	Std.	209.36	D2	Std.	.020	
31	A14	Std.	4321.14	LD15	Std.	.054	RD9	Std.	193.67	A14	Std.	.018	
32	LD8	Std.	4151.44	RD9	Std.	.053	A14	Std.	193.26	RD9	Std.	.017	
33	LD6	Std.	3870.20	RA17	Std.	.046	LA12	Std.	0	LA12	Std.	0	
34	RD9	Std.	3433.04	LA12	Std.	.045	RA17	Std.	0	RA17	Std.	0	0%

(*L*/*R* left/right, *A* angle, *D* distance, *SO* statistical observations, *Avg.* average, *Std.* standard deviation)

25, 50, and 75% (top 25, top 17 and top 8 relevant features are selected, respectively) which were captured from the best recognition rate configuration (60-s window size, 1-sec stepping setting), and the changes in evaluation indices were observed.

Figures 20 and 21 show the test results for the refined feature set extracted from the dataset with the highest recognition rate (60-s window size, 1-s stepping). In Fig. 20, regardless of the feature selection method used, reducing rates of 25 and 50% did not substantially lower the recognition performance, but the training speed significantly increased (Chi-square reaches 2.73×, 4.52×, and IG reaches 2.60× and 4.53×, separately. See as Fig. 21). However, when the reducing rate reached 75%, recognition performance substantially decreased, especially in the case of TEs (Chi-square reached 77.97%; IG reached 57.97%). Also, rather than improving training speed performance, reducing rate

Table 7 The ranking of relevant features—with 60-s window size

#	Step size: 1 s						Step size: 5 s						Reduced by
	Chi-square			Information Gain			Chi-square			Information Gain			
	Name	SO	Score	Name	SO	Score	Name	SO	Score	Name	SO	Score	
1	RA11	Avg.	32056.76	D2	Avg.	.230	RA11	Avg.	3382.33	D2	Avg.	.160	
2	D1	Avg.	24893.33	A14	Avg.	.224	LA10	Avg.	2707.11	A14	Avg.	.159	
3	LA10	Avg.	24318.75	RA11	Avg.	.217	D2	Avg.	1945.30	RA11	Avg.	.142	
4	A13	Std.	18947.85	LD6	Avg.	.208	A14	Avg.	1699.55	LD6	Avg.	.140	
5	D2	Avg.	15215.62	LA10	Avg.	.198	D5	Avg.	1674.26	A13	Avg.	.135	
6	LA10	Std.	14901.22	RD7	Avg.	.196	A13	Avg.	1578.44	LA10	Avg.	.132	
7	D5	Avg.	14884.83	A13	Avg.	.196	LD8	Avg.	1473.90	RD4	Avg.	.127	
8	A14	Avg.	13540.52	D5	Avg.	.186	D1	Avg.	1453.85	LD3	Avg.	.126	75%
9	LD3	Std.	13176.10	RD9	Avg.	.183	RD4	Avg.	1446.40	RD7	Avg.	.124	
10	LD6	Avg.	13036.16	LD15	Avg.	.182	LD6	Avg.	1410.09	LD15	Avg.	.123	
11	LA12	Avg.	12713.67	RD4	Avg.	.179	A13	Std.	1371.47	D5	Avg.	.122	
12	A13	Avg.	12698.15	RD16	Avg.	.179	LD3	Avg.	1366.25	LD8	Avg.	.115	
13	LD8	Avg.	12602.49	LA10	Std.	.179	LD3	Std.	1272.30	RD16	Avg.	.092	
14	RA17	Avg.	12299.63	LD8	Avg.	.178	LD15	Avg.	1246.14	LA10	Std.	.092	
15	LD8	Std.	12174.02	LD3	Avg.	.174	RD7	Avg.	1177.73	RA11	Std.	.091	
16	RD16	Avg.	12103.58	D1	Avg.	.171	LA10	Std.	1158.87	RD9	Avg.	.089	
17	RD9	Avg.	12078.02	A13	Std.	.168	RA11	Std.	1136.29	LD8	Std.	.082	50%
18	RD7	Avg.	12056.51	RA11	Std.	.163	RD4	Std.	1117.63	D1	Std.	.078	
19	D5	Std.	11990.22	LD8	Std.	.158	RA17	Avg.	1074.15	RA17	Avg.	.077	
20	RD4	Std.	11739.08	D5	Std.	.158	LD8	Std.	1020.07	RD4	Std.	.077	
21	LD15	Avg.	11459.00	RD7	Std.	.155	RD9	Avg.	960.64	D1	Avg.	.076	
22	RD4	Avg.	10893.37	LA12	Avg.	.151	LA12	Avg.	953.80	LD3	Std.	.071	
23	RA11	Std.	10723.73	LD3	Std.	.149	RD16	Avg.	937.46	RD7	Std.	.070	
24	LD3	Avg.	10554.41	LD6	Std.	.148	D5	Std.	872.08	LA12	Avg.	.069	
25	RD7	Std.	10516.57	RA17	Avg.	.146	D1	Std.	829.70	RD16	Std.	.068	25%
26	D1	Std.	10507.60	RD4	Std.	.143	RD7	Std.	806.32	D5	Std.	.062	
27	LD6	Std.	10107.95	D1	Std.	.142	RD16	Std.	792.28	LD6	Std.	.061	
28	LD15	Std.	9794.46	RD16	Std.	.134	LD6	Std.	739.94	A14	Std.	.056	
29	RD16	Std.	9702.84	LD15	Std.	.132	A14	Std.	700.76	RD9	Std.	.056	
30	A14	Std.	9071.89	RD9	Std.	.126	LD15	Std.	660.53	LD15	Std.	.052	
31	D2	Std.	8807.48	A14	Std.	.125	RD9	Std.	659.50	A13	Std.	.051	
32	LA12	Std.	8487.44	LA12	Std.	.117	D2	Std.	443.31	D2	Std.	.042	
33	RA17	Std.	8260.37	D2	Std.	.103	LA12	Std.	398.67	LA12	Std.	.032	
34	RD9	Std.	8243.86	RA17	Std.	.107	RA17	Std.	392.63	RA17	Std.	.032	0%

(*L*/*R* left/right, *A* angle, *D* distance, *SO* statistical observations, *Avg.* average, *Std.* standard deviation)

datasets reached only 2.80x and 2.41x by Chi-square and IG, separately.

Another objective of this study was to investigate which facial features result from the emotion transitions and basic emotion recognitions. Therefore, the datasets were divided according to emotion categories (BEs and TEs), and each feature was evaluated separately using the IG and Chi-square from the dataset with the highest recognition rate (60-s window size, 1-s step settings). Table 8 shows the results for the refined feature set.

5.3 The results of the empirical experiment

Figures 22 and 23 and Table 9 show the results for the empirical experiment. In Fig. 22, when the emotion regulation involved in learning process, the negative behaviors

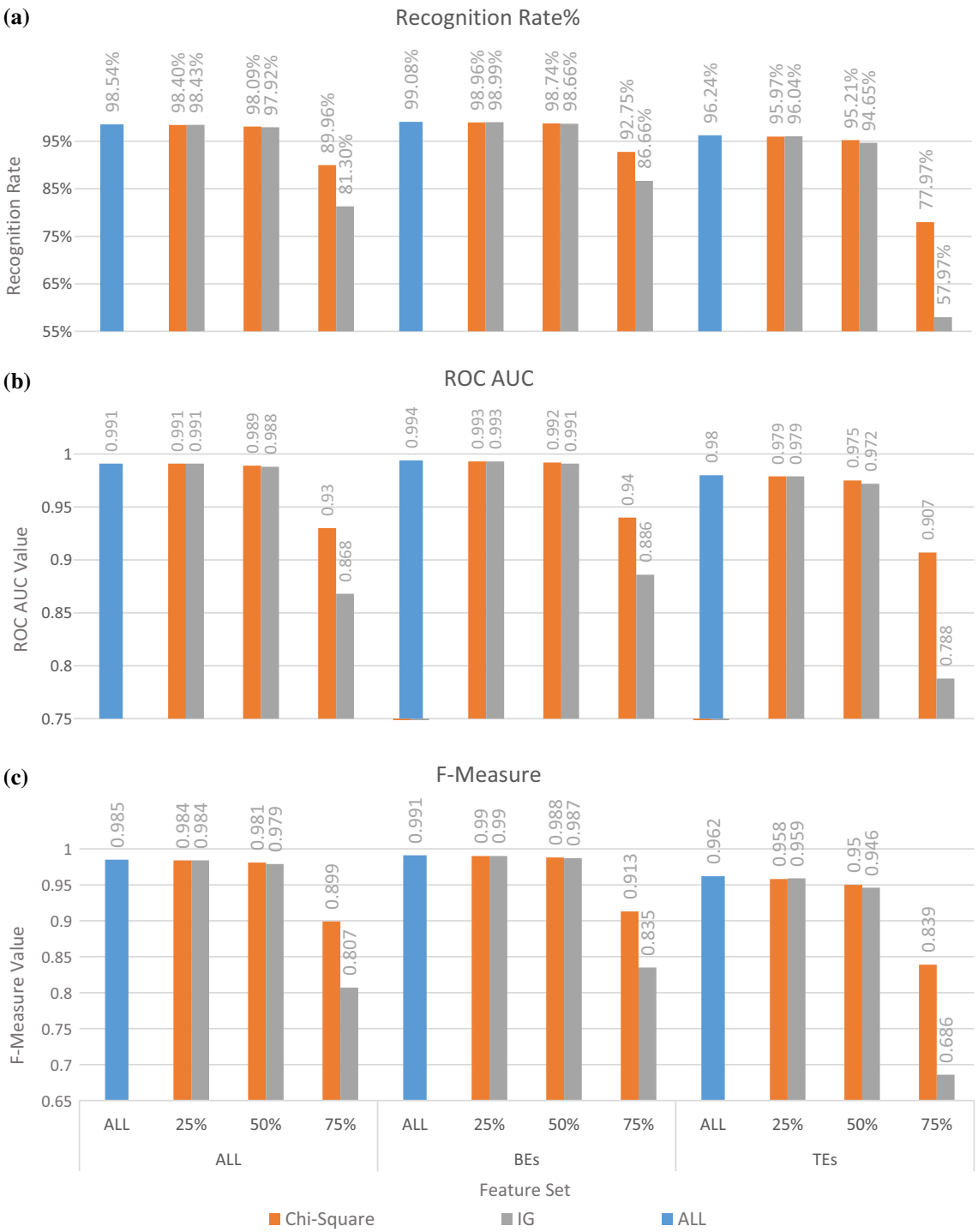


Fig. 20 **a** Average recognition rate of the test results for refined feature set, **b** ROC AUC value of the test results for refined feature set, **c** F-measure value of the test results for refined feature set

were significantly decreased. Also, in Fig. 23, learning performance was improved. In Table 9, the analytical results indicate that both indices reached significant levels, which supports the feasibility and effectiveness of the work.

6 Findings and discussion

The proposed method for emotion recognition with emotion transition detection comprised five parts: sliding window sampling, automated labeling, feature extraction, classifier construction, and the emotion elicitation experiment for col-

Table 8 The ranking of relevant features (divided by emotion categories)

#	Basic emotions (BEs)						Transitional emotions (TEs)					
	Chi-square			Information Gain			Chi-square			Information Gain		
	Name	SO	Score	Name	SO	Score	Name	SO	Score	Name	SO	Score
1	**D2**	**Avg.**	**8337.63**	**D2**	**Avg.**	**.172**	**RA11**	**Avg.**	**11986.45**	**LD6**	**Avg.**	**.552**
2	**RA11**	**Avg.**	**7625.55**	**A14**	**Avg.**	**.163**	**D1**	**Avg.**	**11514.45**	**LD15**	**Avg.**	**.515**
3	**A14**	**Avg.**	**7457.82**	**RD4**	**Avg.**	**.133**	**LA10**	**Avg.**	**11224.57**	**RA11**	**Avg.**	**.512**
4	**LA10**	**Avg.**	**6948.63**	**RA11**	**Avg.**	**.131**	**A13**	**Std.**	**9133.82**	**LA12**	**Avg.**	**.487**
5	**RD4**	**Avg.**	**6372.24**	**A13**	**Avg.**	**.131**	**LD6**	**Avg.**	**7255.03**	**A13**	**Std.**	**.480**
6	LA10	Std.	6159.45	LA10	Avg.	.126	D5	Std.	7007.60	D5	Std.	.478
7	A13	Avg.	5909.57	LD6	Avg.	.126	LD8	Avg.	6829.94	LD8	Avg.	.475
8	LD3	Avg.	5736.04	RD7	Avg.	.124	LD15	Avg.	6756.52	D5	Avg.	.470
9	LD6	Avg.	5678.79	LD3	Avg.	.123	LA12	Avg.	6744.64	RD9	Avg.	.469
10	RA11	Std.	5307.92	LA10	Std.	.113	RA17	Avg.	6405.45	D1	Avg.	.469
11	RD7	Avg.	5221.76	RD9	Avg.	.109	D5	Avg.	6081.10	RD16	Avg.	.462
12	LD8	Std.	5033.08	RA11	Std.	.105	RD9	Avg.	5945.10	LA10	Avg.	.462
13	RD9	Avg.	4980.29	RD16	Avg.	.102	RD16	Avg.	5921.51	RA17	Avg.	.445
14	D5	Avg.	4919.24	D5	Avg.	.099	LA10	Std.	5816.59	LD6	Std.	.433
15	RD16	Avg.	4590.26	LD15	Avg.	.094	LD15	Std.	5412.17	D2	Avg.	.432
16	D1	Avg.	4201.93	LD8	Avg.	.094	LD3	Std.	5386.15	A14	Avg.	.427
17	LD8	Avg.	4082.49	LD8	Std.	.092	D2	Avg.	5285.95	RD7	Avg.	.426
18	A13	Std.	4080.84	RD7	Std.	.087	RD4	Std.	5189.50	LA10	Std.	.416
19	RD7	Std.	4047.68	D1	Avg.	.087	LD6	Std.	5154.95	RA11	Std.	.404
20	LD15	Avg.	3950.97	A13	Std.	.087	LD8	Std.	5154.83	RD4	Std.	.397
21	RD16	Std.	3815.03	RD16	Std.	.078	RD7	Avg.	5121.29	RD7	Std.	.394
22	LA12	Avg.	3464.69	LD6	Std.	.076	RA11	Std.	4913.98	LD8	Std.	.393
23	LD6	Std.	3460.49	RD9	Std.	.072	RD16	Std.	4730.99	LD15	Std.	.390
24	RD9	Std.	3453.23	D1	Std.	.068	A14	Avg.	4598.88	LD3	Std.	.385
25	RA17	Avg.	3120.15	LA12	Avg.	.066	RD7	Std.	4543.83	A13	Avg.	.366
26	LD15	Std.	3113.29	LD15	Std.	.062	D1	Std.	4496.92	RD4	Avg.	.364
27	RD4	Std.	3049.33	D2	Std.	.061	RD9	Std.	4435.17	RD16	Std.	.357
28	D2	Std.	3004.08	RA17	Avg.	.060	RA17	Std.	4316.91	LD3	Avg.	.353
29	LD3	Std.	2921.91	RD4	Std.	.058	LA12	Std.	4234.60	D1	Std.	.348
30	D1	Std.	2871.69	D5	Std.	.058	RD4	Avg.	4201.02	RA17	Std.	.340
31	D5	Std.	2745.69	LD3	Std.	.056	A13	Avg.	4126.70	LA12	Std.	.339
32	A14	Std.	2480.93	A14	Std.	.055	LD3	Avg.	4104.18	RD9	Std.	.333
33	RA17	Std.	2338.06	RA17	Std.	.047	A14	Std.	4043.31	A14	Std.	.329
34	LA12	Std.	2201.80	LA12	Std.	.046	D2	Std.	3626.91	D2	Std.	.294

(*L/R* left/right, *A* angle, *D* distance, *SO* statistical observations, *Avg.* average, *Std.* standard deviation)

lecting emotion-related facial signals to train classifier. Based on results of all fifteen subjects, recognition effectiveness indices were acceptable (ROC AUC > = 0.7). In contrast with previous reports suggesting that the emotional facial features of HFA students are distinguishable from those of normal students (Bieberich and Morgan 2004; Czapinski and Bryson 2003; Yirmiya et al. 1989), the analytical results showed that some recognizable emotional characteristics were common among the fifteen HFA students.

Second, compared with the 5-s stepping sliding window, the 1-s stepping sliding window delivered high-resolution signals and substantially improved the recognition of emotions, especially TEs. The larger sliding window size also delivered more signal ingredients and improved recognition performance (Figs. 14, 15, 16). The analytical results showed that capturing instantaneous changes in facial features through the sliding window with a high sampling rate and larger sample size was beneficial with regard to recognizing transitional emotions. However, there was no significant difference found in feature selection.

Third, as compared to the BPNN, the proposed method showed better discriminability and time savings in the train-

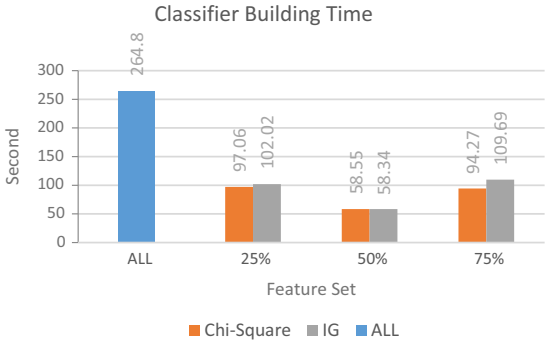


Fig. 21 Classifier building time of the test results for refined feature set

ing process (Fig. 19), especially in the subject-independent test stage. This advantage makes the proposed emotion recognition mechanism easier to deploy in an e-learning platform.

Fourth, the skewed distribution of elicitation results in this experiment indicated that emotions cannot be elicited at an average frequency. The main causes of skewed emotion elicitation were as follows: (i) based on moral and ethical considerations, indicators should limit the strength of elicited negative emotions. (ii) In some students with insufficient learning motivation, elicitation is ineffective. (iii) Due to the presence of parents during the experiment, students may have repressed their negative emotions. Nevertheless, the evaluation results for emotion recognition showed acceptable performance in all categories of emotion.

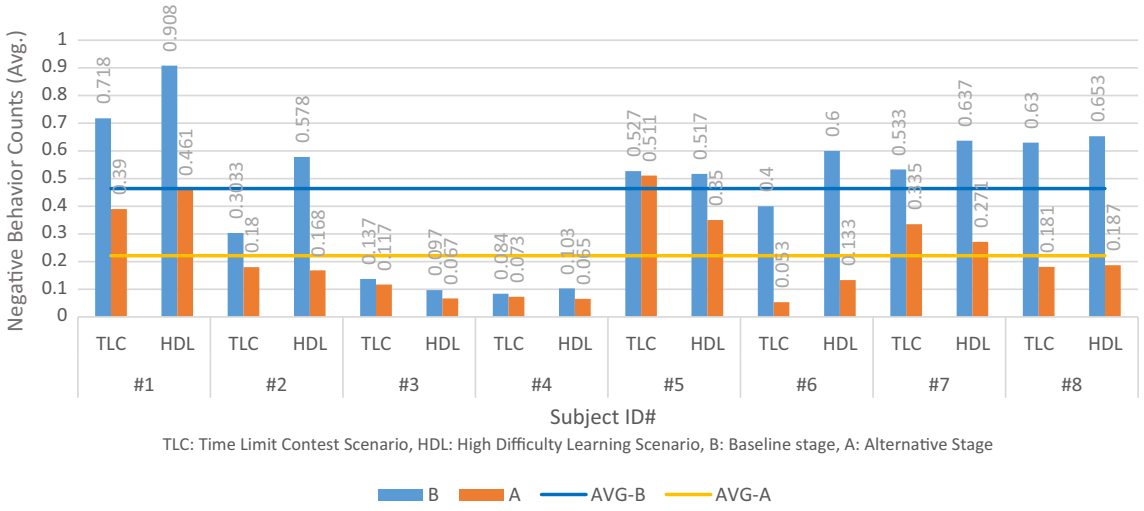


Fig. 22 Negative behavior observation in learning process for eight subjects

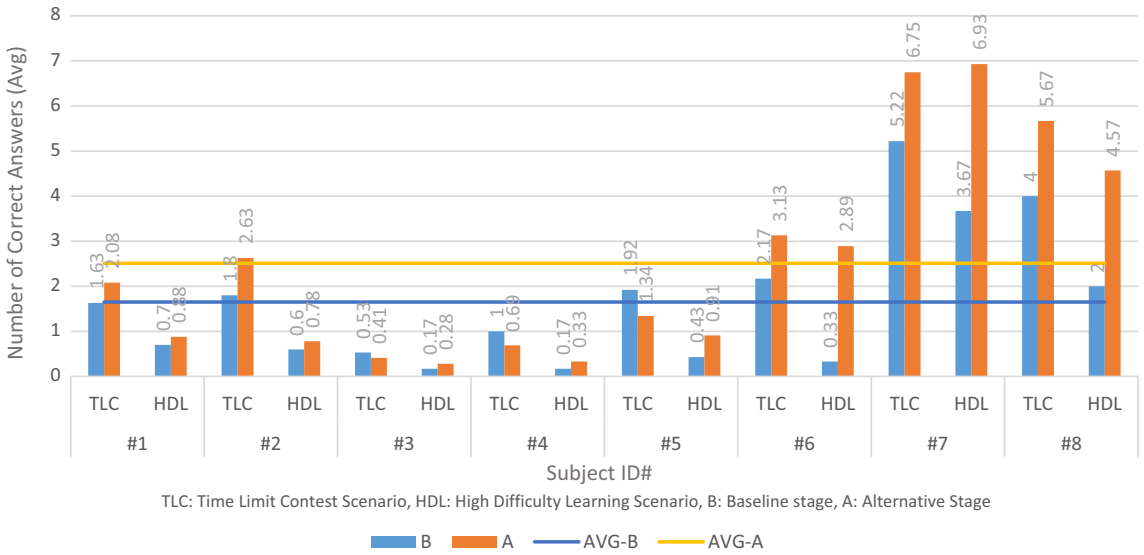


Fig. 23 Math learning performance result for eight subjects

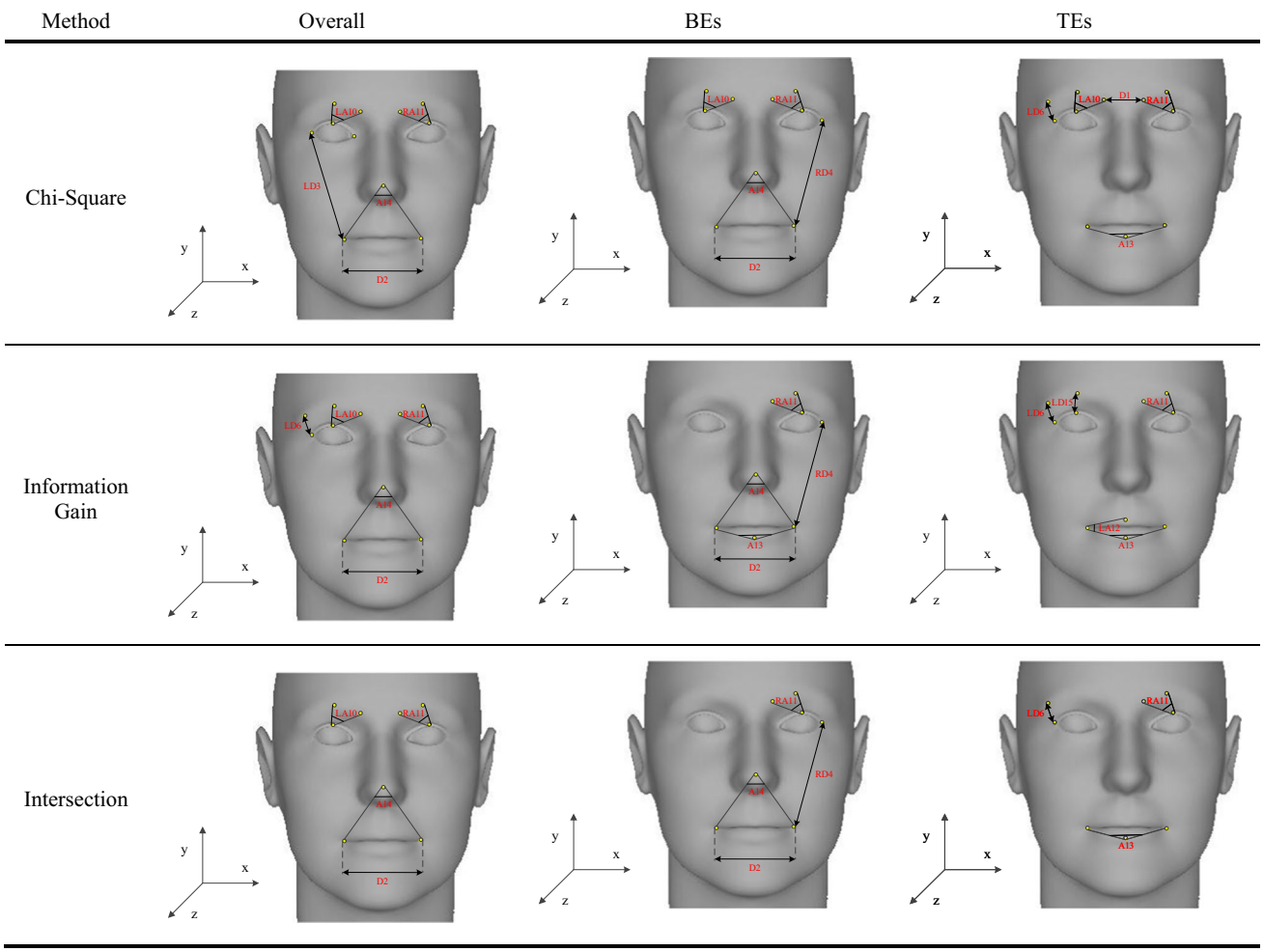


Fig. 24 The top 5 locations of important facial features

Table 9 The results for the empirical experiment

Observation index	p value
Negative behavior rate	1.526e−05 ***
Learning performance	0.0045 **

* $p < .050$; ** $p < .001$; *** $p < .0001$

Fifth, because of the high resolution of signals resampled from the 1-sec stepping sliding window, the IG and Chi-square demonstrated the capability to reduce the dimension of the datasets and training time (Fig. 20). However, when the reducing rate reached 75%, the recognition results significantly decreased.

Sixth, Fig. 24 shows the distribution of the five most important features. (The importance of ranking result is shown in Table 8.) Notably, eyebrows, lids, and mouth size (includes width and height) are important features in recognition. However, in TEs detection, the eyelid-related features and distance between eyebrows (D1) and the size of eye openings (LD15, LD6, RA11) were found to be more important than mouth-related features and also to be different from recognizing BEs. The distribution analysis results showed that transition emotions can be recognized by eye-related features. In contrast, to recognize basic emotions, the emotion recognition cues should take into consideration both mouth-related features and mouth-eye distances (RD4).

Seventh, from the analytical results for the empirical experiment, both emotion regulation effectiveness and mathematics learning performance were improved. The observation indices, negative behavior rate, and learning performance all reached significant levels ($p = 1.526e{-}05$*** in the index for negative behavior rate, $p = 0.0045$** in the learning performance index, see Table 9). The result proved the effectiveness and feasibility of this work.

7 Conclusion and future works

In this study, SVM-based facial emotion recognition with transition detection is proposed. Observing HFA students, who have unique distinctiveness with regard to facial features

as proven by psychologists and educationalists, this study achieved a 98.54% recognition rate, with a .991 ROC AUCs value. For BE recognition, the proposed approach achieved a 99.08% recognition rate, with .994 ROC AUC, and 96.24% with .980 in TEs recognition. Using feature selection to refine the feature sets significantly improved classifier building performance but with some loss of accuracy. Information Gain was used to choose the top 50% of the features and achieved at least a 98.09% overall recognition rate, with 4.52x acceleration in regard to classifier building time. As compared to the BPNN, the proposed approach demonstrated better discriminability and performance. Also, the analytical results for the empirical experiment provided sufficient support of using emotion recognition with regulation in an e-learning environment. This study confirmed the good capability and reliability of the proposed method of detecting emotion transition in emotion recognition and identified the key features used to recognize emotions, including basic emotions and transitional emotions. Of course, some limitations of the method deserve comment. First, additional samples are needed since the facial features elicited from fifteen HFA students may not be representative of all HFA students. The results, therefore, may be limited in generalizing to a larger population. The use of a larger sample size in future experiments will also help researchers to improve the reliability of emotion recognition and will improve the robustness of the results in regard to all HFA student populations.

Second, in practice, HFA students have intense emotional reactions not only in learning, but also in everyday life. Hence, further experiments are needed to elicit emotions in different scenarios. The benefits of performing the emotion elicitation experiment in different scenarios include (i) the potential discovery of new evidence about relations between emotion transition and facial features that may improve accuracy in identifying emotions and (ii) will allow the recognition of their emotions in situations other than learning. Through cooperating with emotional adaptation, ubiquitous services related to the emotional health care of HFA students will become possible.

Third, regarding the skewed elicitation problem, better results can be achieved if (i) the students do not know the experiment is being monitored, and (ii) various experimental environments can be developed instead of test-based learning. Hence, the scenarios used to elicit emotions should be modified to meet these requirements and to improve the effective of elicitation. For example, the experimental environment could be set up in a classroom instead of in a simulated environment.

A final concern is the contrast between the emotional facial features of typical students and those of HFA students. By performing the emotion elicitation experiment in subjects other than HFA students, this study not only obtained evidence of the relation between emotional changes and facial features according to different subject types, but also confirmed the commonality and portability of the proposed method and thus expanded its potential applications to humanoid robots, intelligent agents (Becker et al. 2004), etc.

Compliance with ethical standards

Conflict of Interest Hui-Chan Chu declares that she has no conflict of interest. William Wei-Jen Tsai declares that he has no conflict of interest. Min-Ju Liao declares that she has no conflict of interest. Yuh-Min Chen declares that he has no conflict of interest.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. This article does not contain any studies with animals performed by any of the authors. Informed consent was obtained from all individual participants included in the study.

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